

# **Operational Performance Evaluation of Amman BRT Route 99**

*Using TCQSM-Based Level of Service Measures*

**Ahmad Alkhawaldeh**

*Prepared in collaboration with the Greater Amman Municipality Public Transportation  
Department*

*Supervised by Eng. Mohammed Al-Sarihin and Eng. Adel Al-Daboubi*

# Abstract

## Abstract

This paper evaluates the operational performance of Amman BRT Route 99 using TCQSM-based Level of Service (LOS) measures and operational data from the Amman Vision Intelligent Transportation System (ITS). The analysis draws on a seven-day dataset and examines schedule adherence, headway reliability, expected passenger waiting time, dwell time, operating speed, and passenger comfort to assess service quality across time periods and route directions.

The results indicate that Route 99 achieves an overall composite LOS of C; however, this aggregate rating conceals major reliability deficiencies. While high service frequency supports the overall LOS, the corridor experiences severe headway irregularity, recurrent early running, unstable dwell times, and crowding during the PM peak. These conditions reduce service predictability and degrade the passenger experience despite the route's relatively strong frequency performance.

The study shows that the main operational weaknesses are linked to brittle schedule design, insufficient resilience in running time allocation, and stop-level friction associated with passenger activity. The paper therefore provides an applied basis for schedule recalibration, dwell time reduction strategies, and more targeted peak-period capacity management. As one of the first detailed TCQSM-based operational evaluations of Route 99, the study offers a practical framework for performance-based improvement of the wider Amman BRT network.

# 1.0 Introduction

## 1.1 Problem Statement: The Infrastructure – Operation Gap

Amman has grown rapidly over recent decades, and this growth has placed sustained pressure on the city's transport system. Congestion, car dependence, and uneven access to reliable public mobility have become defining features of daily travel, while Jordan's public transport sector continues to face structural challenges related to service quality, integration, and operational effectiveness. The Amman Bus Rapid Transit (BRT) system was introduced in this context as a major effort to improve urban mobility and support a shift toward more efficient public transport.

The performance of a BRT system, however, is not determined by infrastructure alone. Dedicated lanes, upgraded stations, and modern vehicles create the physical conditions for high-quality service, but these advantages can be undermined when day-to-day operations become irregular or poorly matched to passenger demand. Preliminary inspection of the operational data for Route 99 indicated that reliability may be a meaningful concern on the corridor, with signs of inconsistent headways and uneven schedule performance. This raises the possibility of an *infrastructure–operation gap*: a system that possesses the physical components of BRT while still underperforming at the operational level. Because reliability is central to passenger trust and to the long-term role of mass transit in shifting travel away from private vehicles, this gap warrants a rigorous, evidence-based examination rather than descriptive assessment alone.

## 1.2 Research Aim

This study evaluates the operational performance of Amman BRT Route 99 using a quantitative, diagnostic approach grounded in the *Transit Capacity and Quality of Service Manual* (TCQSM). The framework and its Level of Service (LOS) measures are introduced in detail in Chapter 2; here it is sufficient to note that they translate raw operational data into standardized indicators of service quality, allowing performance to be assessed consistently across stops, time periods, and directions of travel. The evaluation focuses on four dimensions: service reliability, operating speed, capacity, and passenger comfort. Within these, particular attention is given to schedule adherence, headway regularity, passenger waiting conditions, dwell time behavior, and loading patterns. The aim is not only to assign performance grades, but to identify the specific mechanisms through which service quality is being lost on the corridor.

## 1.3 Study Objectives and Contribution

The study pursues four objectives:

- To benchmark the operational reliability of Route 99 by measuring schedule and headway adherence across peak and off-peak periods using TCQSM-based LOS thresholds.

- To examine passenger loading conditions and comfort levels and identify periods and segments where demand places strain on service quality.
- To evaluate dwell time patterns and operating speed in order to distinguish between infrastructure-related delay and operational inefficiency.
- To develop evidence-based recommendations for improving service stability, schedule design, and peak-period management on the corridor.

In doing so, the paper contributes the first detailed stop-level, TCQSM-based operational diagnosis of an Amman BRT corridor and offers a directional, time-of-day analytical method that can be replicated across the wider network.

## 2.0 Literature Review

### 2.1 Bus Rapid Transit: Definition, Promise, and Operational Reality

Bus Rapid Transit has emerged over the past two decades as one of the most widely adopted public transport strategies in fast-growing cities, particularly where rail investment is constrained by cost or implementation time. The concept combines the operational flexibility of buses with several features traditionally associated with rail: dedicated right-of-way, intersection priority, upgraded stations with platform-level boarding, off-board fare collection, and intelligent transport systems for vehicle management (Levinson et al., 2003; ITDP, 2016). When these elements are implemented together, BRT can deliver high passenger throughput, competitive travel times, and a service quality approaching that of light rail at a fraction of the capital cost.

In practice, however, BRT systems vary widely in how fully they incorporate these features, and the gap between "BRT-branded" service and high-performance BRT can be substantial. Hidalgo and Gutiérrez (2013), reviewing more than thirty systems globally, observed that operational underperformance, rather than infrastructure deficiency, is often the main reason BRT corridors fail to deliver their promised benefits. The ITDP BRT Standard (2016) likewise distinguishes between corridors based on the quality and consistency of their operations, not only their physical design. This distinction is particularly relevant for routes that combine dedicated and mixed-traffic segments, where infrastructure alone cannot guarantee reliable service. Route 99 in Amman, which operates partly in dedicated lanes and partly in mixed traffic, sits squarely within this operational reality.

### 2.2 The TCQSM and the Measurement of Transit Service Quality

The analytical framework adopted in this study is the *Transit Capacity and Quality of Service Manual*, third edition (Kittelson & Associates et al., 2013), commonly referred to as the TCQSM. The manual is the most widely used reference for transit performance measurement in North America and has been applied internationally as a standardized basis for evaluating service quality. Its central methodological contribution is the systematic separation of operator-oriented indicators (such as vehicle supply and capacity) from passenger-oriented indicators (such as waiting time, travel time reliability, and onboard comfort), and the expression of these indicators through Level of Service (LOS) categories ranging from A to F.

The LOS framework allows technical operational data to be translated into interpretable performance grades, which makes it well suited to applied evaluation work. It also makes performance comparable across stops, time periods, routes, and even systems, provided the underlying measures are computed consistently. For these reasons, the TCQSM has become a common reference point in BRT and bus corridor evaluations, including in cities where it serves as a benchmark rather than a regulatory standard (Diab and El-Geneidy, 2013; Tirachini et al., 2014). The framework does have limitations — its thresholds were calibrated primarily on North American operating contexts, and it does not prescribe a method for combining dimensions into a single composite index — but as a structured basis for diagnostic analysis it remains the most appropriate tool for the present study.

## **2.3 Reliability, Headway Regularity, and the Passenger Experience of Frequent Service**

Reliability is consistently identified in the empirical literature as one of the most important determinants of transit service quality, often ranking above raw travel time in studies of passenger satisfaction (Carrel et al., 2013). For high-frequency services in particular, reliability is most usefully understood not as adherence to a published timetable, but as the regularity of spacing between consecutive vehicles. When headways are short, passengers tend to arrive at stops randomly rather than to a schedule, and their effective waiting time depends on how evenly buses are distributed in time rather than on whether each individual run is "on time" (Furth and Muller, 2006).

The mechanism most commonly associated with the breakdown of headway regularity is bus bunching, first formalized by Newell and Potts (1964) and developed in the modern operations literature by Daganzo (2009). The dynamic is well established: when one bus falls behind schedule, it accumulates additional passengers, dwells longer at subsequent stops, and falls further behind, while the bus following it gains time, picks up fewer passengers, and closes the gap. Without active intervention, the result is pairs or clusters of buses operating close together, separated by long service gaps. From the passenger perspective, this raises expected waiting time well above half the mean headway and produces uneven crowding across vehicles, even when the timetabled frequency is unchanged (Daganzo and Pilachowski, 2011).

A consistent finding in the empirical literature is that headway irregularity tends to amplify along a route rather than remain stable. Strathman et al. (2002), using automatic vehicle location data from Portland, showed that the coefficient of variation of headways grows systematically with distance from the origin terminal, and that this growth is a stronger predictor of passenger waiting penalty than any single delay event. This pattern has direct implications for how reliability should be evaluated: corridor-level averages can mask substantial downstream deterioration, and stop-level analysis is therefore essential. Several authors have proposed control strategies to address bunching, ranging from holding at timing points (Furth and Muller, 2007) to dynamic headway-based control using real-time vehicle data (Daganzo, 2009; Bartholdi and Eisenstein, 2012). The relevance of these strategies depends, in each case, on whether the underlying instability is diagnosed correctly, which itself requires the kind of stop-level reliability analysis adopted in this study.

## **2.4 Dwell Time, Stop-Level Friction, and the Propagation of Delay**

Dwell time, the period during which a vehicle remains stopped at a station while passenger activity and associated processes take place, is widely recognized as one of the principal sources of variability in bus and BRT operations. The TCRP literature on dwell time (Kittelson & Associates et al., 2013; Levinson, 1983) identifies several determinants beyond passenger volume itself: boarding process (front-door pay-as-you-board versus all-door boarding), fare collection method, door configuration, vertical alignment between platform and vehicle floor, and the presence of wheelchair or stroller boarding events. Each of these factors can shift the average time required to process a given number of passengers, and several can also increase its variability.

The operational significance of dwell time extends beyond its direct contribution to travel time. Because dwell variability propagates through a route, a longer-than-average stop produces a fuller bus at the next stop, which then dwells longer, dwell time is a central mechanism through which schedule and headway irregularity emerge in practice. Tirachini (2013) and Fletcher and El-Geneidy (2013) have shown empirically that interventions targeting boarding efficiency, such as off-board fare collection and all-door boarding, produce reliability gains that exceed the direct dwell-time savings, because they reduce the variance of stop-level processing time and therefore weaken the bunching mechanism.

Methodologically, dwell time is also useful as a diagnostic lens. Examining average dwell time alongside passenger boarding activity allows analysts to distinguish between stops where delay is primarily demand-driven (high boardings, strong correlation between boardings and dwell) and stops where delay is primarily process-driven or environment-driven (high dwell times that persist even when few or no passengers board). This decomposition is particularly valuable in corridors where targeted interventions are needed, since the appropriate response, passenger-processing improvements versus operational or procedural changes, depends on the underlying cause. The dwell-time analysis presented in Chapter 4 builds directly on this distinction.

## **2.5 The Amman Context and the Position of the Present Study**

The Amman BRT system was developed within this context as part of a broader effort to modernize the city's public transport network through both infrastructure investment and operational reform. Published evaluations of the system's operational performance, particularly at the stop level, remain limited. This study therefore contributes to a small but growing literature on BRT performance in the region, while applying an internationally established analytical framework that allows the findings to be situated alongside comparable evaluations elsewhere.

Published evaluations of BRT operational performance in the Middle East and North Africa region remain limited, and detailed stop-level operational analyses of corridors in Jordan are particularly scarce. The bulk of the international BRT performance literature has been produced in Latin American, North American, and East Asian contexts, with a smaller body of work emerging on developing-country systems elsewhere. The present study contributes to this regional gap by applying an internationally established analytical framework to a corridor that has not previously been examined at this level of detail, allowing the findings to be situated alongside comparable evaluations in other contexts.

## **2.6 Summary and Position of the Present Study**

Taken together, the literature reviewed above establishes three points that frame the analytical approach of this study. First, BRT performance is determined by the interaction between infrastructure and operations, and operational shortcomings can substantially erode the benefits of high-quality physical design. Second, in high-frequency service the most consequential reliability measure is headway regularity rather than schedule adherence, and headway irregularity tends to amplify downstream through mechanisms such as bus bunching. Third,

dwell time is both a contributor to unreliability and a diagnostic instrument, since separating its demand-driven and process-driven components helps locate the operational sources of corridor-level performance problems.

The present study applies these principles to Amman BRT Route 99 using the TCQSM framework and stop-level operational data. In doing so, it aims to provide a structured, evidence-based diagnosis of where and how service quality is being lost on the corridor, and to support targeted operational improvements rather than generic recommendations.

## References

- Bartholdi, J. J., and Eisenstein, D. D. (2012). A self-coordinating bus route to resist bus bunching. *Transportation Research Part B: Methodological*, 46(4), 481–491.
- Carrel, A., Halvorsen, A., and Walker, J. L. (2013). Passengers' perception of and behavioral adaptation to unreliability in public transportation. *Transportation Research Record*, 2351, 153–162.
- Daganzo, C. F. (2009). A headway-based approach to eliminate bus bunching: Systematic analysis and comparisons. *Transportation Research Part B: Methodological*, 43(10), 913–921.
- Daganzo, C. F., and Pilachowski, J. (2011). Reducing bunching with bus-to-bus cooperation. *Transportation Research Part B: Methodological*, 45(1), 267–277.
- Diab, E. I., and El-Geneidy, A. M. (2013). Variation in bus transit service: Understanding the impacts of various improvement strategies on transit service reliability. *Public Transport*, 4(3), 209–231.
- Fletcher, G., and El-Geneidy, A. (2013). Effects of fare payment types and crowding on dwell time: A fine-grained analysis. *Transportation Research Record*, 2351, 124–132.
- Furth, P. G., and Muller, T. H. J. (2006). Service reliability and hidden waiting time: Insights from automatic vehicle location data. *Transportation Research Record*, 1955, 79–87.
- Furth, P. G., and Muller, T. H. J. (2007). Service reliability and optimal running time schedules. *Transportation Research Record*, 2034, 55–61.
- Hidalgo, D., and Gutiérrez, L. (2013). BRT and BHLS around the world: Explosive growth, large positive impacts and many issues outstanding. *Research in Transportation Economics*, 39(1), 8–13.
- Imam, R. (2014). Measuring public transport performance using AHP in Amman, Jordan. *International Journal of Engineering Research and Technology*, 3(6), 423–429.
- Institute for Transportation and Development Policy (ITDP). (2016). *The BRT Standard, 2016 Edition*. New York: ITDP.



Kittelsohn & Associates, Parsons Brinckerhoff, KFH Group, Texas A&M Transportation Institute, and Arup. (2013). *Transit Capacity and Quality of Service Manual, Third Edition* (TCRP Report 165). Washington, DC: Transportation Research Board.

Kittelsohn & Associates, KFH Group, Parsons Brinckerhoff Quade & Douglass, and Hunter-Zaworski, K. (2003). *Transit Capacity and Quality of Service Manual, 2nd Edition* (TCRP Report 100). Washington, DC: Transportation Research Board.

Levinson, H. S. (1983). Analyzing transit travel time performance. *Transportation Research Record*, 915, 1–6.

Levinson, H. S., Zimmerman, S., Clinger, J., Rutherford, S., Smith, R. L., Cracknell, J., and Soberman, R. (2003). *Bus Rapid Transit, Volume 1: Case Studies in Bus Rapid Transit* (TCRP Report 90). Washington, DC: Transportation Research Board.

Mahmoud, M. S., Habib, K. N., and Shalaby, A. (2014). Survey of cross-regional intermodal passenger travel (SCRIPT): A joint stated-preference survey for measuring multimodal cross-regional travel demand. *Transportation*, 41(5), 1167–1185.

Newell, G. F., and Potts, R. B. (1964). Maintaining a bus schedule. In *Proceedings of the 2nd Australian Road Research Board Conference*, vol. 2, 388–393.

Strathman, J. G., Kimpel, T. J., Dueker, K. J., Gerhart, R. L., and Callas, S. (2002). Evaluation of transit operations: Data applications of Tri-Met's automated bus dispatching system. *Transportation*, 29(3), 321–345.

Tirachini, A. (2013). Bus dwell time: The effect of different fare collection systems, bus floor level and age of passengers. *Transportmetrica A: Transport Science*, 9(1), 28–49.

Tirachini, A., Hensher, D. A., and Rose, J. M. (2014). Multimodal pricing and optimal design of urban public transport: The interplay between traffic congestion and bus crowding. *Transportation Research Part B: Methodological*, 61, 33–54.

### 3.0 Methodology

This study adopts a quantitative, diagnostic approach to evaluate the operational performance of Amman BRT Route 99 using operational data from the Amman Vision Intelligent Transportation System (ITS). The analysis applies the *Transit Capacity and Quality of Service Manual*, third edition (Kittelson et al., 2013) as the principal evaluation framework, translating raw operational records into standardized Level of Service (LOS) measures across schedule reliability, headway regularity, dwell-time behavior, operational efficiency, and passenger comfort. The methodology combines route-level and stop-level analyses to examine how service performs across different times of day and in both directions of travel.

#### 3.1 Study Area

The study focuses on Route 99 of the Amman Bus Rapid Transit network, which operates between Jordan Museum Terminal and Swaileh Terminal over a corridor of approximately 17 kilometers. The route functions as a major high-capacity trunk corridor and combines dedicated BRT lanes with mixed-traffic segments, making it a representative case for evaluating operational performance under varying physical and traffic conditions. The corridor connects key activity nodes, including the University of Jordan, Jordan University Hospital, Sport City, and central commercial and government zones.

The analysis covers both operating directions:

- **Direction 100990 (Outbound: Swaileh → Jordan Museum)** — 26 stops.

| Stop Sequence | Stop ID | Stop Name                 |
|---------------|---------|---------------------------|
| 1             | 10612   | Swaileh Terminal          |
| 2             | 10613   | Hesham ben Alaas ST       |
| 3             | 11675   | Yajouz ST                 |
| 4             | 10616   | Alrubane ST               |
| 5             | 11671   | Ahmad Altarawneh ST       |
| 6             | 10617   | University of Jordan      |
| 7             | 10618   | Agriculture Collage       |
| 8             | 10619   | J.U Hospital              |
| 9             | 10620   | Mahmoud Alkiswa ST        |
| 10            | 10621   | Abd Al lateef Abu Qura ST |
| 11            | 10622   | Alsahafeh Tunnel          |
| 12            | 10623   | Mukhtar Mall              |
| 13            | 10625   | Sport City Terminal       |
| 14            | 10626   | Abd Alhameed Sharaf ST    |
| 15            | 10627   | Omar ben Abd Alazeez ST   |
| 16            | 10628   | King Abdullah Gardens     |
| 17            | 10640   | Wadi Saqra Int.           |

|    |       |                        |
|----|-------|------------------------|
| 18 | 10629 | Mecca ST               |
| 19 | 10630 | Mohammad Janah ST      |
| 20 | 10631 | Fawzi Alqawqji ST      |
| 21 | 10632 | Mohammad Bdair ST      |
| 22 | 10633 | Barda ST               |
| 23 | 10641 | Al-Muhajireen          |
| 24 | 10635 | Hussain Culture Center |
| 25 | 10636 | Amman Municipality     |
| 26 | 10637 | Jordan Museum Terminal |

- **Direction 100991 (Inbound: Jordan Museum → Swaileh) — 25 stops.**

| Stop Sequence | Stop ID | Stop Name                 |
|---------------|---------|---------------------------|
| 1             | 10643   | Jordan Museum Terminal    |
| 2             | 10636   | Amman Municipality        |
| 3             | 10635   | Hussain Culture Center    |
| 4             | 10634   | Prs. Basmah ST            |
| 5             | 10633   | Barda ST                  |
| 6             | 10632   | Mohammad Bdair ST         |
| 7             | 10631   | Fawzi Alqawqji ST         |
| 8             | 10630   | Mohammad Janah ST         |
| 9             | 10629   | Mecca ST                  |
| 10            | 10628   | King Abdullah Gardens     |
| 11            | 10627   | Omar ben Abd Alazeez ST   |
| 12            | 10626   | Abd Alhameed Sharaf ST    |
| 13            | 10625   | Sport City Terminal       |
| 14            | 10623   | Mukhtar Mall              |
| 15            | 10622   | Alsahafeh Tunnel          |
| 16            | 10621   | Abd Al lateef Abu Qura ST |
| 17            | 10620   | Mahmoud Alkiswani ST      |
| 18            | 10619   | J.U Hospital              |
| 19            | 10618   | Agriculture Collage       |
| 20            | 10617   | University of Jordan      |
| 21            | 10615   | Ahmad Altarawneh ST       |
| 22            | 10616   | Alrubane ST               |
| 23            | 10614   | Yajouz ST                 |
| 24            | 10613   | Hesham ben Alaas ST       |
| 25            | 10642   | Swaileh Terminal          |

For readability, the directional shorthand "Outbound" and "Inbound" is used throughout the remainder of the paper alongside the system direction codes.

### 3.2 Data Source and Study Period

The analysis is based on operational data obtained from the Amman Vision ITS, the central platform used by the operator to manage and monitor BRT service. The system records GPS-based vehicle locations, scheduled and actual arrival and departure times at each stop, vehicle identifiers, and passenger boarding counts. Data were collected over a continuous seven-day period from 1 to 7 August 2025.

The August window was selected to capture typical operating conditions during a continuous, uninterrupted week of service. Operationally, this period falls within the University of Jordan's summer semester, during which the campus continues to function but at lower student volumes than during the first and second semesters. Demand on Route 99 during the study period therefore represented an intermediate, rather than maximum, demand condition for the corridor. This is acknowledged as a contextual factor when interpreting load-factor results in Chapter 4: peak academic-term demand would be expected to push load factors higher than the values reported here, particularly at university-adjacent stops.

To support time-of-day analysis, observations were stratified into four operating periods reflecting variations in demand and traffic conditions on the corridor:

| Time Period | Hours       | Description                 |
|-------------|-------------|-----------------------------|
| AM Peak     | 07:00–10:00 | High-demand morning hours   |
| AM Off-Peak | 10:00–14:00 | Moderate midday demand      |
| PM Peak     | 14:00–18:00 | Afternoon congestion period |
| PM Off-Peak | 18:00–22:00 | Lower-demand evening hours  |

### 3.3 Schedule Baseline Normalization

A central methodological challenge in the analysis was the misalignment between the published service schedule recorded in the ITS and the actual dispatch behavior observed in operation. Published terminal departure times were frequently not honored: vehicles departed earlier or later than the scheduled time, sometimes by several minutes, and substitutions in vehicle assignment occurred without a corresponding update to the ITS schedule. In effect, day-to-day operations followed an informal dispatch logic rather than the timetable encoded in the system. Several factors are likely to contribute to this pattern, including driver discretion at the terminal, block scheduling constraints, vehicle availability, and the absence of a strictly enforced dispatch protocol.

Using the original published schedule as the reference for on-time performance would have conflated two different problems: structural irregularity in dispatch, which originates at the terminal before service begins, and operational deviation in route, which arises from running-time and stop-level conditions during the trip itself. Because the focus of this study is corridor-level operational performance, a schedule baseline normalization procedure was applied:

1. For each individual trip, the actual observed departure time at the origin terminal was treated as the on-time reference point, replacing the published departure.
2. Scheduled arrival times at all downstream stops were then recomputed by adding the planned stop-to-stop travel increments from the ITS schedule to this normalized departure.
3. Schedule deviation at every downstream stop was measured against this trip-specific normalized baseline rather than against the original published times.

The effect of this procedure is that the schedule adherence analysis evaluates whether each trip remained consistent with its own intended block-time profile during operation, isolating in-route performance from upstream dispatch irregularity. This isolation is essential to the interpretation of the on-time performance results in Chapter 4. In particular, the recurrent early running observed in the Outbound direction reflects operational behavior *given* a normalized starting point, that is, structural slack in the block time itself, and is not an artifact of buses leaving terminals ahead of schedule.

### 3.4 Data Preparation and Trip Sampling

Raw ITS records were processed in several stages. Individual operational records were first reconstructed into complete trips by aligning vehicle identifiers, stop sequences, and timestamps, identifying continuous trip start and end points. Trips with incomplete GPS coverage, missing stop sequences, or broken continuity were excluded from analyses requiring full trip structure, particularly schedule and headway reliability assessment. Trips were then segmented temporally, into the four operating periods defined above, and spatially, into stop-to-stop links to support running-speed and dwell-time analysis at the stop level.

For analyses requiring a controlled and comparable trip sample, six fully complete and uninterrupted trips were selected from each direction for each of the four operating periods, on each of the seven study days. This produced 42 sampled trips per direction per period, and approximately 4,000 stop-level observations across the study window. Trips were drawn consecutively within each period to preserve realistic operational sequences rather than discrete random observations, while excluding trips with missing or corrupted records.

### 3.5 Data Limitations

Three limitations should be noted explicitly.

First, the ITS dataset records boarding events but not passenger alightings. As a result, true onboard occupancy at each stop cannot be reconstructed, and load factor is computed from

total cumulative trip boardings rather than from instantaneous load. Cumulative boardings provide a useful trip-level indicator of vehicle utilization, but they understate the peak segment load that any individual passenger experiences. This means the load-factor LOS grades reported in Chapter 4 should be interpreted as approximate trip-level utilization rather than as direct equivalents to the TCQSM's onboard-load thresholds.

Second, the seven-day observation window is short relative to longer-term temporal evaluations. The extensive cleaning and trip reconstruction required for stop-level analysis limited the feasible observation period for this stage of the research, and the study is therefore framed as a detailed short-term corridor diagnosis rather than a long-term temporal assessment. Future work could extend this analysis across a longer horizon to capture seasonal and recurrent variations.

Third, as noted above, the August period coincides with a summer-semester academic schedule. Demand patterns at university-adjacent stops are likely to be lower than during the first or second semester, and the load-factor results should be read with this in mind.

### **3.6 Performance Measures and LOS Evaluation Framework**

The performance measures applied in this study were drawn directly from the TCQSM (Kittelson et al., 2013), with two adaptations made for the local operating context, which are noted explicitly where they occur. The framework covers four service dimensions: schedule reliability, headway regularity and passenger waiting conditions, dwell-time behavior, and operational efficiency and passenger comfort. A composite LOS index is then constructed from these dimensions to summarize overall corridor performance.

#### **3.6.1 Schedule Adherence**

Schedule adherence was evaluated against the normalized schedule baseline described in Section 3.3, using the standard TCQSM on-time window of  $-1$  to  $+5$  minutes. Arrivals more than one minute earlier than the normalized scheduled time were classified as early; arrivals more than five minutes later were classified as late; intermediate arrivals were classified as on-time. The  $-1$  minute lower bound reflects the passenger consequence of early running on frequent service: a bus arriving even a minute ahead of schedule may cause a waiting passenger to miss the trip.

Three indicators were computed:

**Mean Deviation (MD):** the average difference between actual and normalized scheduled arrival time.

$$MD = \frac{1}{N} \sum_{i=1}^N (T_{a,i} - T_{s,i})$$

where  $T_{a,i}$  and  $T_{s,i}$  are the actual and normalized scheduled arrival times for observation  $i$ , and  $N$  is the number of observations. Negative values indicate early running; positive values indicate late running.

**Standard Deviation (SD)** of arrival deviation:

$$SD = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (T_{a,i} - T_{s,i} - MD)^2}$$

This captures the variability in schedule performance and therefore reflects predictability from the passenger perspective.

**On-time performance (OTP)** was defined as the percentage of arrivals falling within the -1 to +5 minute window. The OTP thresholds applied in this study are consistent with the LOS bands presented in the TCQSM 2nd Edition (Kittelson et al., 2003), which were derived from a survey of more than 80 U.S. transit agencies' on-time performance standards. The 2nd Edition bands were retained for this analysis because they reflect agency-level operational thresholds that remain widely used in practice and provide a more graduated band structure for an applied corridor evaluation than the narrower bands introduced in the 3rd Edition. The OTP value at each stop was used to assign an LOS grade following the TCQSM thresholds for fixed-route service reliability:

| LOS | % On-Time | Quality Description      | Color Code  |
|-----|-----------|--------------------------|-------------|
| A   | ≥ 95%     | Excellent reliability    | Green       |
| B   | 90–94%    | Good reliability         | Light Green |
| C   | 75–89%    | Fair reliability         | Yellow      |
| D   | 60–74%    | Marginal reliability     | Orange      |
| E   | 50–59%    | Poor reliability         | Red         |
| F   | < 50%     | Unacceptable reliability | Dark Red    |

### 3.6.2 Headway Reliability and Passenger Waiting Time

Because passenger experience on frequent service depends more on the regularity of bus spacing than on adherence to a published timetable (Furth and Muller, 2006), headway-based measures were applied at the stop level. Three indicators were used.

**Mean Observed Headway** ( $\bar{H}$ ) is the average time interval between successive bus arrivals at a stop, providing the effective service frequency under observed operating conditions:

$$\bar{H} = \frac{1}{N} \sum_{i=1}^N h_i$$

**Coefficient of Variation of Headways** ( $CV_h$ ) measures regularity of spacing and is the principal indicator of headway reliability:

$$CV_h = \frac{\sigma_h}{\bar{H}}$$

where  $\sigma_h$  is the standard deviation of observed headways. Higher values indicate more unstable spacing and a stronger likelihood of bunching and gapping.

**Expected Passenger Waiting Time** ( $E[W]$ ) translates headway irregularity into a passenger-facing measure. Under random passenger arrivals, the TCQSM expresses expected waiting time as:

$$E[W] = \frac{\bar{H}}{2} (1 + CV_h^2)$$

When service is perfectly regular,  $E[W]$  approaches  $\bar{H}/2$ . As  $CV_h$  rises, expected waiting time grows disproportionately, capturing the passenger penalty associated with irregular spacing.

LOS grades for the three measures were assigned using the TCQSM thresholds:

**Service Frequency (mean headway):**

| LOS | Mean Headway (min) | Description                               | Color Code  |
|-----|--------------------|-------------------------------------------|-------------|
| A   | $\leq 10$          | Passengers do not need a schedule         | Green       |
| B   | 10.1–14            | Frequent; passengers may consult schedule | Light Green |



|   |         |                                               |          |
|---|---------|-----------------------------------------------|----------|
| C | 14.1–20 | Maximum desirable wait if a vehicle is missed | Yellow   |
| D | 20.1–30 | Service unattractive to choice riders         | Orange   |
| E | 30.1–60 | Service available about once per hour         | Red      |
| F | > 60    | Less than once per hour                       | Dark Red |

### Headway Reliability ( $CV_h$ ):

| LOS | $CV_h$      | Description       | Color Code  |
|-----|-------------|-------------------|-------------|
| A   | 0.00–0.21   | Very reliable     | Green       |
| B   | 0.22–0.30   | Reliable          | Light Green |
| C   | 0.31–0.39   | Some irregularity | Yellow      |
| D   | 0.40–0.52   | Irregular         | Orange      |
| E   | 0.53–0.74   | Very irregular    | Red         |
| F   | $\geq 0.75$ | Highly unreliable | Dark Red    |

The expected-waiting-time grade was assigned by mapping  $E[W]$  onto the same passenger-perception bands implied by the mean-headway thresholds, since under perfectly regular service  $E[W] = \bar{H}/2$ . This produced the LOS bands used for  $E[W]$  in Chapter 4.

### 3.6.3 Dwell Time

Dwell time refers to the duration during which a bus remains stopped at a station while passenger activity and associated stop processes take place. Because dwell variability propagates along a route and is a recognized contributor to headway irregularity (Levinson, 1983; Tirachini, 2013), dwell time was analyzed both as a performance outcome and as a diagnostic indicator. The following measures were computed at every stop, in each direction, for each of the four operating periods:

- Average dwell time.
- Average dwell time on stops with recorded passenger boardings.
- Average dwell time on stops without recorded passenger boardings.
- Standard deviation and coefficient of variation of dwell time.
- The coefficient of determination ( $R^2$ ) between the number of boarding passengers and dwell time at the stop.

Comparing dwell time on with-passenger and without-passenger stops, together with the strength of the boardings–dwell relationship ( $R^2$ ), allows stops to be classified diagnostically. A stop where dwell rises substantially with boardings, with a high  $R^2$ , is consistent with demand-driven delay. A stop with persistent dwell time even when no boardings are recorded, and a low  $R^2$ , is more consistent with process-driven or environment-driven delay (such as procedural overhead, platform layout, or operator behavior). This decomposition is used in Chapter 4 to distinguish stop-level bottlenecks by likely cause.

It should be noted that, because the ITS records boarding events but not alightings, dwell time classified as "without passengers" reflects the absence of recorded boardings only, and may still include passenger alighting activity. This caveat applies throughout the dwell-time analysis.

A coefficient-of-variation interpretation table was used as a secondary diagnostic for dwell-time stability:

| CV of Dwell Time | Stability                       |
|------------------|---------------------------------|
| $\leq 0.20$      | Stable                          |
| 0.21–0.40        | Moderately variable             |
| $> 0.40$         | High variability / inconsistent |

### 3.6.4 Operational Efficiency and Passenger Comfort

Three trip-level measures were used to evaluate operational efficiency and passenger comfort.

**Load Factor (LF)** was computed as the ratio of total trip boardings to vehicle capacity:

$$LF = \frac{B}{C} \times 100\%$$

where B is the cumulative number of boardings on the trip and C is the seated-plus-standing capacity of the vehicle assigned to that trip. Vehicle capacity was assigned according to bus type, identified through the vehicle code prefix in the ITS records. Three vehicle types operated on the corridor during the study period:

| Vehicle Code Prefix | Bus Type     | Capacity (passengers) |
|---------------------|--------------|-----------------------|
| V                   | Vectio C Bus | 44                    |
| K                   | Kent C Bus   | 62                    |
| Z                   | Kinglong Bus | 62                    |

As noted in Section 3.5, this load factor represents trip-level utilization rather than instantaneous onboard load, and underestimates peak segment crowding.

**Operating Speed (S)** was computed from corridor length and elapsed trip time:

$$S = \frac{60 L}{T}$$

where L is the corridor length in kilometers, T is the elapsed trip time in minutes, and S is in km/h.

**Buffer Time Index (BTI)** was used as a trip-level travel-time reliability measure, expressing the additional proportion of mean travel time a passenger would need to allow to arrive on time with 95% probability:

$$BTI = \frac{T_{95} - \bar{T}}{\bar{T}}$$

where  $T_{95}$  is the 95th-percentile trip time and  $\bar{T}$  is the mean trip time across the period.

LOS thresholds for load factor and operating speed were applied as follows. The load-factor thresholds used here adapt the TCQSM passenger-comfort framing for an applied trip-level cumulative measure, and the operating-speed thresholds were adopted from BRT performance benchmarks consistent with TCQSM-derived speed bands (Kittelson et al., 2013; ITDP, 2016):

**Load Factor:**

| LOS | Load Factor | Passenger Experience          | Color Code  |
|-----|-------------|-------------------------------|-------------|
| A   | ≤ 50%       | Seats available for everyone  | Green       |
| B   | 51–70%      | Comfortable standing possible | Light Green |
| C   | 71–90%      | Crowded; some standing        | Yellow      |
| D   | 91–100%     | Very crowded                  | Orange      |
| E   | 101–120%    | Full; uncomfortable           | Red         |
| F   | > 120%      | Overcrowded                   | Dark Red    |

**Operating Speed:**

| LOS | Speed (km/h) | Description    | Color Code  |
|-----|--------------|----------------|-------------|
| A   | ≥ 35         | Very fast      | Green       |
| B   | 25–34        | Fast           | Light Green |
| C   | 20–24        | Moderate       | Yellow      |
| D   | 15–19        | Slow           | Orange      |
| E   | 10–14        | Very slow      | Red         |
| F   | < 10         | Extremely slow | Dark Red    |

### 3.6.5 Composite LOS Index

To summarize corridor performance across dimensions, a weighted composite LOS index was computed for each direction. The TCQSM does not prescribe a single composite weighting scheme. The weights used here were chosen for this study and are interpreted in Chapter 4 as a summary measure that complements, rather than replaces, the dimensional analysis.

The index integrates four components: reliability (R), frequency (F), operating speed (S), and passenger comfort (C). LOS grades for each component were converted to a numerical scale in which A=5, B=4, C=3, D=2, E=1, F=0, and the composite was computed as a weighted average:

$$\text{LOS overall} = 0.30R + 0.30F + 0.20S + 0.20C$$

The weights reflect the relative emphasis placed on the two passenger-facing dimensions encountered at the point of access, reliability and frequency, weighted at 30% each, and the two in-vehicle dimensions, speed and comfort, weighted at 20% each. These weights are discussed openly as a methodological choice in Section 4.6.

Two further methodological choices should be stated explicitly. First, the reliability component is computed by averaging stop-level  $CV_h$ -based grades across all stops and all four operating periods within a direction, producing a single reliability score per direction. Frequency, speed, and comfort are computed by averaging their respective period-level grades within a direction. This aggregation rule treats stops and periods symmetrically and is consistent across components.

Second, schedule adherence is *not* included as a separate component of the composite. This is a deliberate choice intended to avoid double-counting with the headway-reliability dimension: in high-frequency service, schedule deviation and spacing irregularity are driven by overlapping operational mechanisms, and reviewers of the composite should treat headway reliability as the dimension carrying that information. The schedule adherence findings reported in Section 4.2

therefore inform, but do not directly enter, the composite. The implications and limitations of this design, including the structural role of frequency in setting a high floor on the composite when service is supplied densely, are discussed in Section 4.6.4.

The resulting composite score is converted back to a letter grade using equal-interval boundaries on the 0–5 scale.

## 4.0 Results and Analysis

This chapter presents the results of the operational performance evaluation for Amman BRT Route 99 using the TCQSM-based measures introduced in Chapter 3. Findings are organized by service dimension and compared across both directions of travel and across the four operating periods defined earlier — AM peak, AM off-peak, PM peak, and PM off-peak. Where useful, an all-day view is also reported to summarize overall corridor performance. For readability, the directional shorthand Outbound (Direction 100990, Swaileh → Jordan Museum) and Inbound (Direction 100991, Jordan Museum → Swaileh) is used throughout the chapter and adopted from first mention onward.

### 4.1 Schedule Adherence (On-Time Performance)

Schedule adherence was evaluated against the normalized schedule baseline defined in Chapter 3 using a tolerance window of  $-1$  to  $+5$  minutes. Arrivals more than one minute earlier than the normalized scheduled time were classified as early, arrivals more than five minutes later were classified as late, and arrivals falling within the window were classified as on-time. The  $-1$  minute lower bound matters because on frequent service an early arrival is not a benign event from the passenger perspective: a bus that arrives ahead of schedule can cause a waiting passenger to miss the trip entirely.

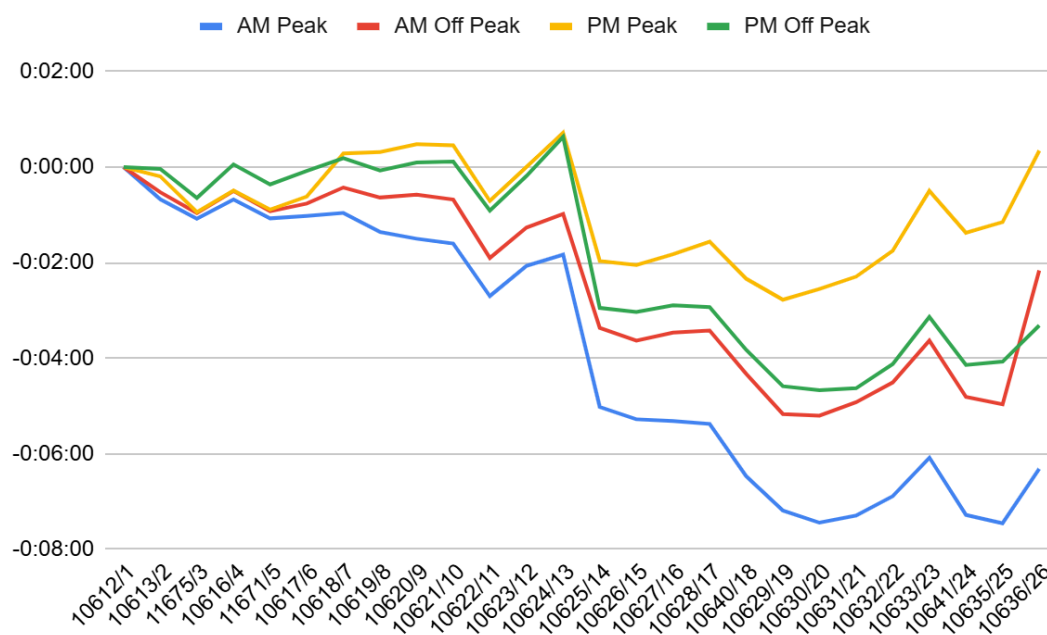
Before turning to the stop-level results, it is worth describing an operating practice observed on Route 99 during the study period, because it shapes how the deviation patterns below should be read. Buses on the corridor do not dwell to a schedule. Drivers stop only to board or alight passengers and otherwise keep moving; they do not hold at timing points to absorb accumulated lead time. This practice prioritizes through-trip speed and reduces in-vehicle delay for boarded passengers, which is not unreasonable in itself. It does, however, interact with the published schedule in a specific way: where the schedule embeds dwell allowances that exceed what drivers actually incur in operation, the difference accumulates along the route and shows up as progressive lead at downstream stops. The deviation results below make this interaction visible.

#### 4.1.1 Mean Deviation (MD)

The mean deviation results show a clear directional asymmetry in how the schedule and corridor operations interact. In the Outbound direction, the dominant pattern is persistent early running across most periods. Mean deviation becomes increasingly negative as buses progress along the route, particularly during the AM peak, AM off-peak, and PM off-peak periods. In the morning periods, buses cross the early-arrival threshold within the first third of the route and continue to gain time, with the largest deviations recorded in the latter half of the corridor. This downstream amplification is consistent with the operating practice described above: when actual dwells fall short of the scheduled allowance, the resulting time savings compound across successive stops rather than being absorbed at any single point. The PM peak is the only period in which this pattern is partially moderated, as downstream congestion offsets some of the accumulated lead near the approach to Jordan Museum. Even then, mid-route performance

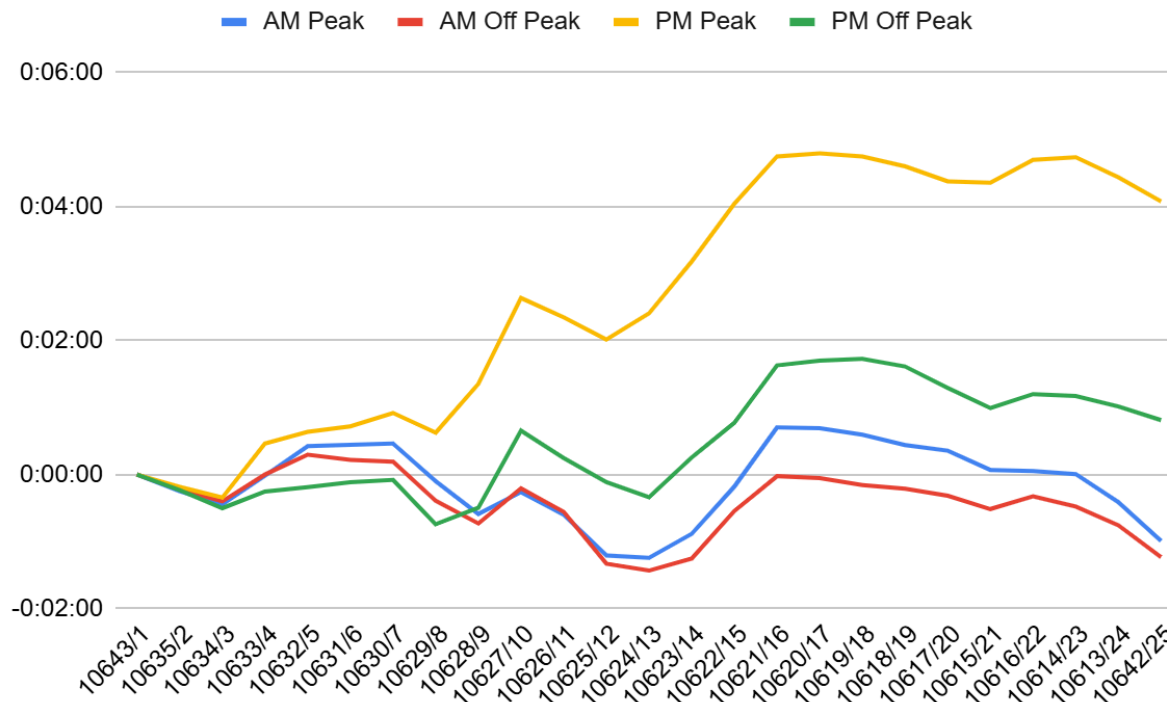
remains unstable, indicating that the near on-time terminal arrival in this period reflects compensating delay rather than reliable adherence across the corridor.

*Line Chart 1. Mean deviation by stop for the Outbound direction across the four time periods.*



In the Inbound direction, the picture is more mixed and the PM peak failure mode is the reverse of Outbound. During the AM peak, AM off-peak, and PM off-peak periods, most stops remain relatively close to the scheduled time, with only limited early running in the mid-route segment and mild deviation toward the terminal. The PM peak, however, shows a clear reversal. Mean deviation shifts from near-zero conditions at the early stops to sustained positive deviation in the second half of the route, indicating systematic lateness rather than excess slack. Lateness becomes pronounced from approximately Stop 10 onward and remains elevated through to the terminal. This pattern is not produced by dwell behavior — Inbound buses operate the same no-dwell practice as Outbound — but by running time itself: evening congestion in the Swaileh-bound direction pushes actual running time above the value embedded in the scheduled block, and the schedule has no mechanism to absorb that additional time.

*Line Chart 2. Mean deviation by stop for the Inbound direction across the four time periods.*



The two directions therefore display different failure modes of the same underlying issue. Outbound is characterized by excess running time relative to what the corridor needs under current operating practice; Inbound is characterized by insufficient running time during the PM peak.

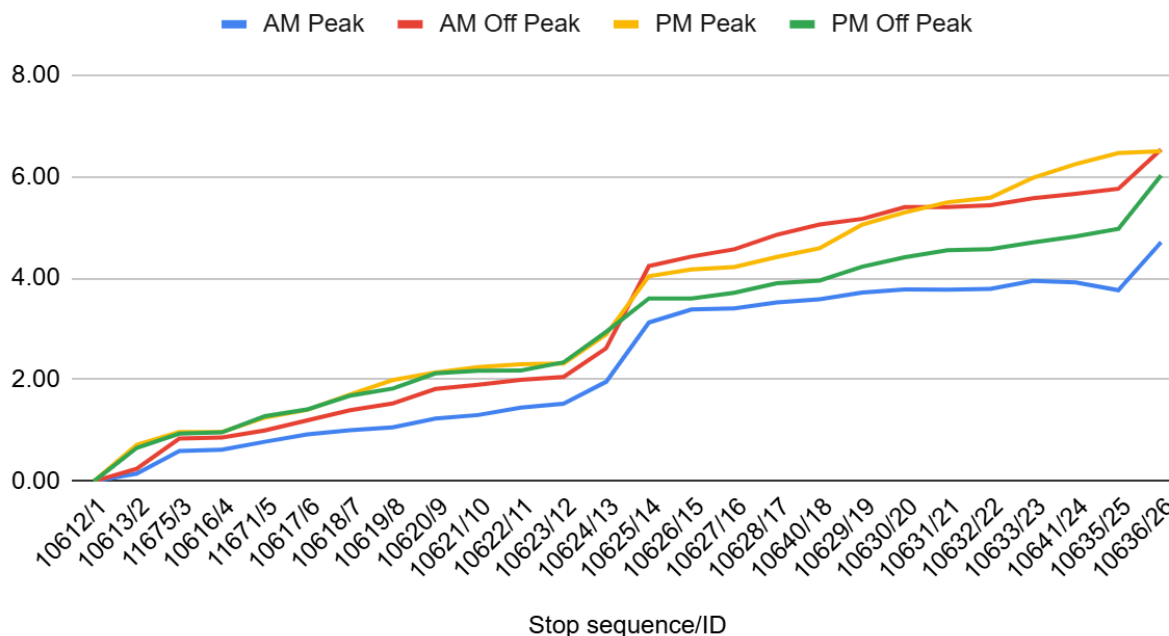
#### 4.1.2 Standard Deviation (SD)

The standard deviation results reinforce these directional patterns by showing that schedule adherence problems are not limited to average deviation but also involve substantial variability in arrival times. In the Outbound direction, variability grows progressively along the route, with downstream stops exhibiting far less predictable arrival performance than the initial segment. Even where the service is consistently early on average, the magnitude of earliness is not stable, different trips consume the available slack at different rates depending on the boarding activity each trip encounters. The downstream growth in standard deviation indicates that the compounding mechanism described above does not produce a uniform lead, but a variable one that further reduces the usefulness of the timetable for passengers.



*Line Chart 3. Standard deviation by stop for the Outbound direction across the four time periods.*

### AM Peak, AM Off Peak, PM Peak and PM Off Peak



In the Inbound direction, variability is comparatively moderate during the AM off-peak and PM off-peak periods but rises sharply during the PM peak, with several downstream stops exceeding five minutes of standard deviation. The combination of systematic lateness and high variability indicates that PM peak Inbound service is both delayed and difficult to predict, reflecting trip-to-trip differences in congestion intensity rather than a uniform schedule problem.



In the Inbound direction, non-peak periods generally perform better in the early part of the route, but reliability still weakens in the latter half of the corridor. During the PM peak, this degradation becomes more severe and is driven primarily by late arrivals rather than early running. The stop-level LOS pattern therefore differs in character from Outbound: where Outbound is dominated by progressive early running, Inbound shows a more context-specific decline tied to evening congestion and insufficient running time. Both directions, however, display substantial schedule adherence deficiencies over large portions of the route beyond the mid-route segment.

*Table 2. Stop-level on-time performance LOS grades for the Inbound direction across the four time periods.*

| Stop /ID    | 1064<br>3/1 | 1063<br>5/2 | 1063<br>4/3 | 1063<br>3/4 | 1063<br>2/5 | 1063<br>1/6 | 1063<br>0/7 | 1062<br>9/8 | 1062<br>8/9 | 1062<br>7/10 | 1062<br>6/11 | 1062<br>5/12 | 1062<br>4/13 | 1062<br>3/14 | 1062<br>2/15 | 1062<br>1/16 | 1062<br>0/17 | 1061<br>9/18 | 1061<br>8/19 | 1061<br>7/20 | 1061<br>5/21 | 1061<br>6/22 | 1061<br>4/23 | 1061<br>3/24 | 1064<br>2/25 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| AM Peak     | A           | A           | B           | C           | B           | B           | B           | C           | E           | E            | E            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            | E            |
| AM Off Peak | A           | A           | A           | B           | B           | B           | B           | C           | E           | D            | D            | F            | F            | E            | E            | D            | D            | D            | E            | E            | E            | E            | E            | F            | F            |
| PM Peak     | A           | A           | A           | A           | B           | B           | B           | C           | D           | D            | D            | D            | F            | E            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            | F            |
| PM Off Peak | A           | A           | A           | D           | D           | C           | C           | F           | E           | D            | D            | F            | E            | F            | F            | F            | E            | E            | F            | F            | E            | E            | F            | F            | F            |

Taken together, the schedule adherence findings suggest that the current operating schedule for Route 99 is too static for the service conditions it is intended to manage. Direction 100990 appears to include excessive running time for most periods outside the PM peak, while Direction 100991 requires additional running time specifically during the PM peak and only modest adjustment outside that window. These results point to the need for direction-specific and time-of-day-specific schedule recalibration rather than a single uniform block-time structure for the entire route.

#### 4.1.4 Synthesis

The schedule adherence findings point to direction-specific and time-dependent reliability failures rather than a single uniform scheduling problem. In Outbound, the dominant issue is persistent early running across most of the service day, indicating that the allocated running time is generally too generous outside the PM peak. In Inbound, the schedule performs reasonably during non-peak periods but becomes inadequate during the PM peak, when buses experience sustained lateness in the second half of the route.

Read against the operating practice described at the start of this section, both patterns are best understood as manifestations of a single underlying mismatch between the published schedule and how the corridor is actually operated. Outbound and Inbound buses both run the same no-dwell practice, holding only to board or alight passengers. In Outbound, that practice combined with a schedule that embeds substantial dwell allowance produces compounding early running. In Inbound during the PM peak, the same practice combined with congestion that exceeds the block's running-time provision produces compounding lateness. One mechanism, the schedule is not calibrated to current operating practice, surfaces as two different failure modes under different operating conditions.

The implication is that a single static block-time structure is not sufficient for this corridor. Recalibration should be direction-specific and time-of-day-specific, built from observed dwell distributions and observed running times rather than from a uniform headway-derived block. In practical terms, this means tightening Outbound block times outside the PM peak to reflect the running times the corridor actually produces, and extending Inbound PM peak block times to cover the congestion-driven running time that the current schedule fails to absorb.

## 4.2 Headway Reliability and Passenger Waiting Time

Headway reliability describes how consistently buses are spaced along the corridor and how that spacing translates into passenger waiting conditions. On high-frequency service, regularity of arrivals is often more important to passengers than strict timetable adherence: irregular spacing produces bunching, gapping, and longer-than-expected waits even when nominal frequency remains high. It also affects how passengers are distributed across vehicles — when service becomes uneven, some buses accumulate heavy loads while following buses operate with lighter ones, producing uneven crowding along the corridor. For these reasons, the analysis combines three linked indicators: mean observed headway, the coefficient of variation of headways ( $CV_h$ ), and expected passenger waiting time ( $E[W]$ ).

### 4.2.1 Mean Actual Headway

Mean actual headway describes the average spacing between successive buses and supports an assessment of service frequency relative to TCQSM thresholds. Across both directions and all four time periods, observed mean headways were short enough to fall within LOS A at every stop. The frequency layer of the service is therefore consistently strong, and the stop-level LOS grade tables for this measure are uniformly A, for that reason, only the underlying headway profiles are reported below.

In the Outbound direction, average headways remained low throughout the day. AM peak headways sat just above 4 minutes, while AM off-peak and PM peak averaged close to 5 minutes for most stops. The PM off-peak produced the widest spacing, generally around 7 minutes, but still well below the 10-minute threshold for LOS A. The all-period average across the route was close to 5 minutes, confirming a consistently high frequency in this direction.

Line Chart 5. Mean actual headway by stop for the Outbound direction across the four time periods.

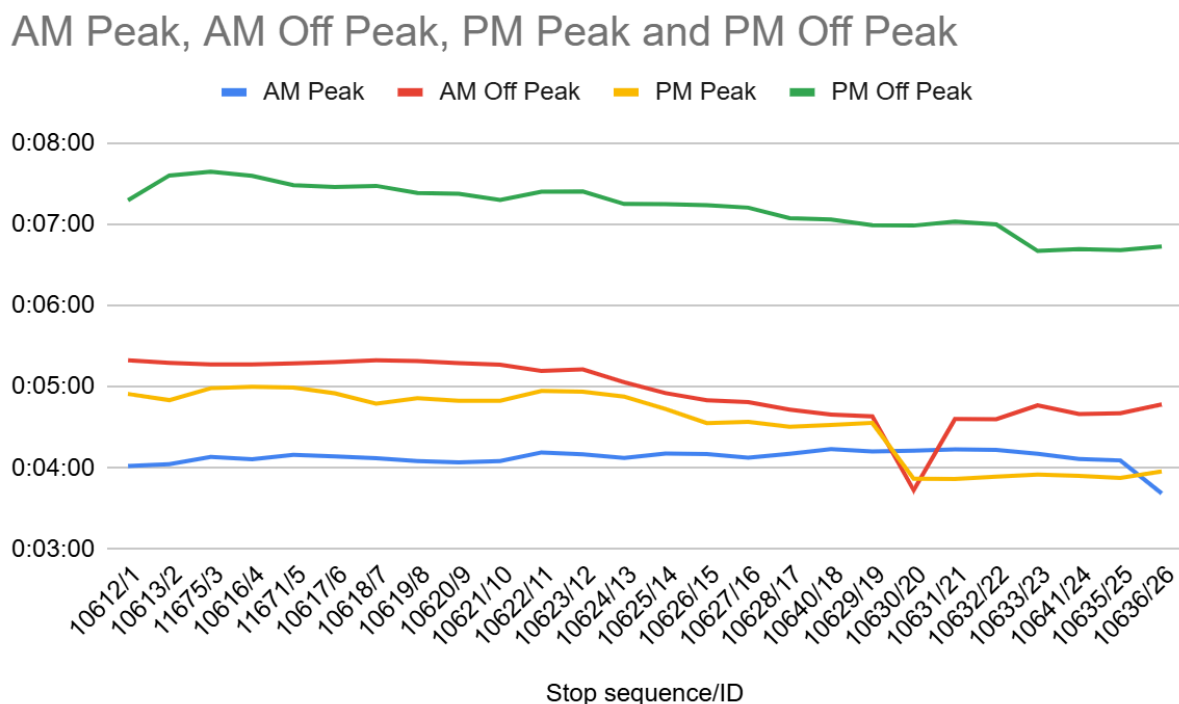


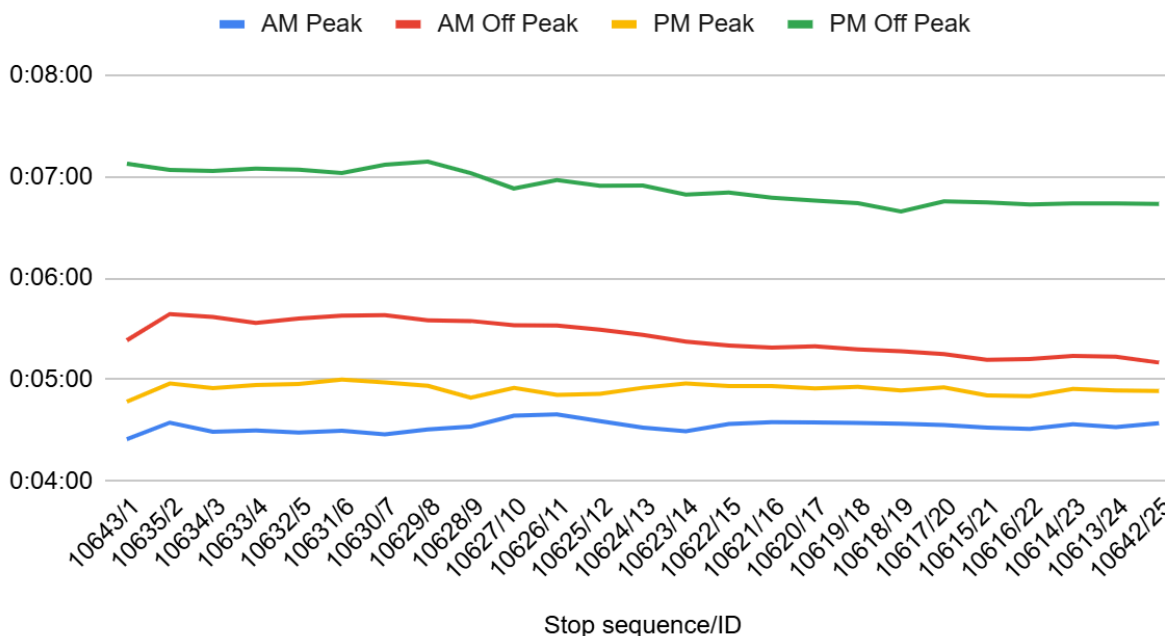
Table 3. Mean actual headway by stop LOS grades for the Outbound direction across the four time periods.

| Stop sequence/ID | 10612/1 | 10613/2 | 11675/3 | 10616/4 | 11671/5 | 10617/6 | 10618/7 | 10619/8 | 10620/9 | 10621/10 | 10622/11 | 10623/12 | 10624/13 | 10625/14 | 10626/15 | 10627/16 | 10628/17 | 10640/18 | 10629/19 | 10630/20 | 10631/21 | 10632/22 | 10633/23 | 10641/24 | 10635/25 | 10636/26 |
|------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| AM Peak          | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| AM Off Peak      | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| PM Peak          | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| PM Off Peak      | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |

The Inbound direction showed a similar pattern. AM peak headways were generally around 4.5 minutes, AM off-peak around 5.5 minutes, and PM peak close to 5 minutes. As in Outbound, PM off-peak headways were the longest at roughly 7 minutes. The all-period average across the route was approximately 5.25 minutes.

*Line Chart 6. Mean actual headway by stop for the Inbound direction across the four time periods.*

### AM Peak, AM Off Peak, PM Peak and PM Off Peak



*Table 4. Mean actual headway by stop LOS grades for the Inbound direction across the four time periods.*

| Stop sequence/ID | 10643/1 | 10635/2 | 10634/3 | 10633/4 | 10632/5 | 10631/6 | 10630/7 | 10629/8 | 10628/9 | 10627/10 | 10626/11 | 10625/12 | 10624/13 | 10623/14 | 10622/15 | 10621/16 | 10620/17 | 10619/18 | 10618/19 | 10617/20 | 10615/21 | 10616/22 | 10614/23 | 10613/24 | 10642/25 |
|------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| AM Peak          | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| AM Off Peak      | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| PM Peak          | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |
| PM Off Peak      | A       | A       | A       | A       | A       | A       | A       | A       | A       | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        | A        |

A comparison of the two directions shows that Route 99 performs strongly in terms of service frequency regardless of direction or time of day. The differences that do exist are relatively modest: Direction 100990 tends to operate with slightly shorter headways in the AM peak, while both directions converge toward longer but still frequent service in the PM off-peak. Overall, the mean headway results indicate that the route's principal operational weakness is not insufficient service supply. On the contrary, buses are deployed at high frequency throughout the day, which means that later reliability problems must be understood primarily in terms of irregular spacing rather than a shortage of vehicles.

#### 4.2.2 Coefficient of Variation of Headways ( $CV_h$ ) and Headway Reliability

While mean headway describes average spacing, reliability depends on how consistently that spacing is maintained. The coefficient of variation of headways was used to assess the degree of irregularity in service spacing along the corridor. Higher  $CV_h$  values indicate greater variability between successive bus arrivals and a stronger likelihood of bunching and gapping, even when average frequency remains high. As an interpretive anchor for the values reported below:  $CV_h$  around 0.30–0.40 indicates noticeable irregularity (LOS C–D); values at or above 0.75 correspond to LOS F, where most buses arrive bunched; and values approaching or exceeding 1.0 indicate that headway variation matches the average spacing itself, operationally, the corridor has lost any meaningful control over bus spacing.

In the Outbound direction, headway reliability was poor across much of the corridor and deteriorated progressively toward the downstream end. During the AM peak,  $CV_h$  began in the range of 0.56 to 0.69 and then rose sharply beyond the mid-route, exceeding 0.80 from approximately Stop 13 onward and reaching 1.20 at the final stop. By the downstream segment, variation in spacing was therefore comparable to — and at the terminus greater than — the average spacing itself. The AM off-peak began from a slightly better starting condition but followed the same pattern of rapid reliability loss, again reaching LOS F from the middle of the route onward. The PM peak produced the most severe values, with  $CV_h$  rising above 1.0 at the later stops and peaking at 1.59 at the terminus. Even during the PM off-peak, when average frequency was lower but still strong, reliability degraded into LOS F near the downstream end. Across all periods, the dominant Outbound pattern is a progressive loss of spacing control as buses move south.



Line Chart 7. Coefficient of variation of headways by stop for the Outbound direction across the four time periods.

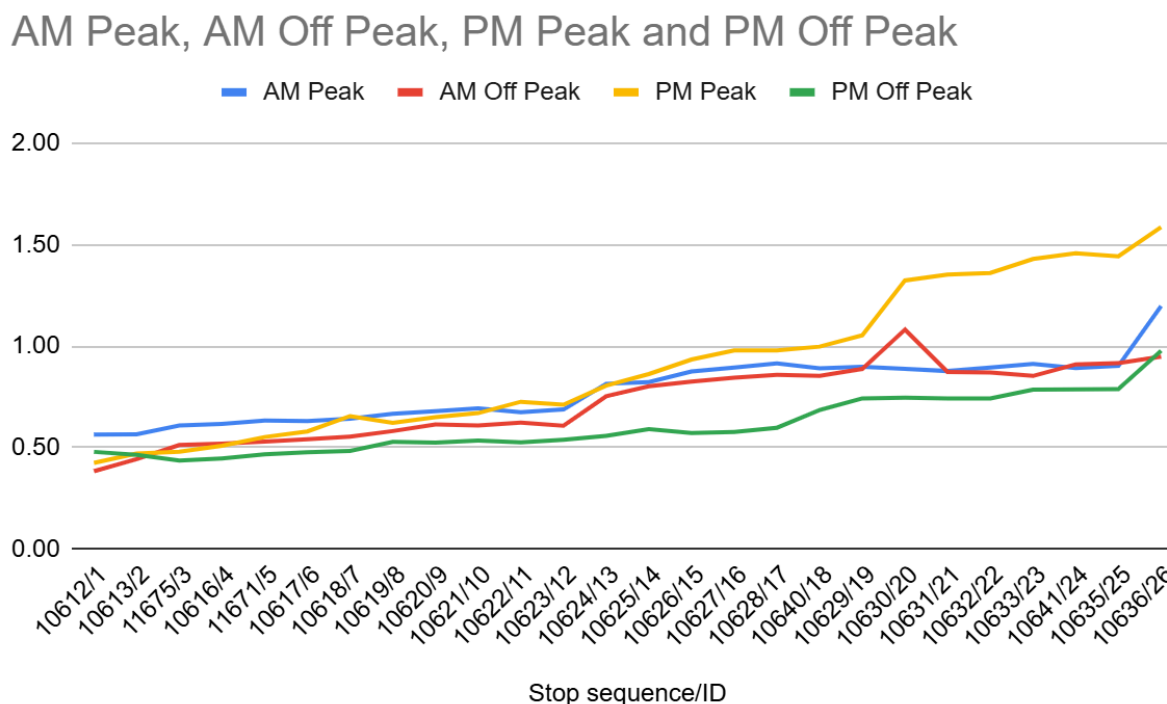


Table 5. Stop-level headway reliability LOS grades for the Outbound direction across the four time periods.

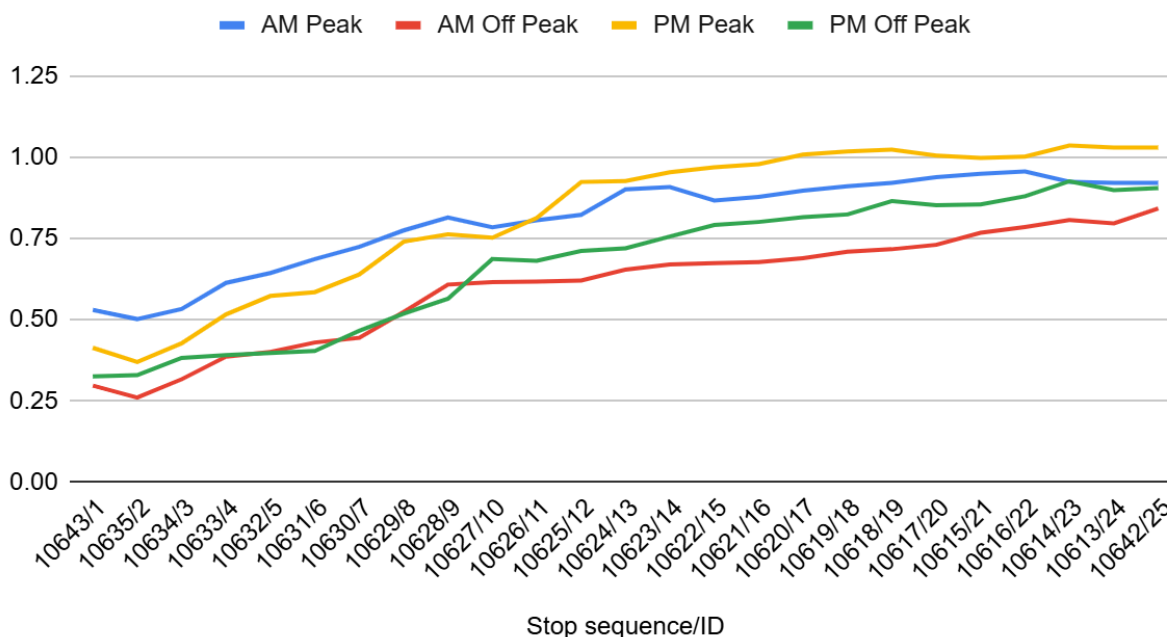
| Stop sequence/ID | 10612/1 | 10613/2 | 11675/3 | 10616/4 | 11671/5 | 10617/6 | 10618/7 | 10619/8 | 10620/9 | 10621/10 | 10622/11 | 10623/12 | 10624/13 | 10625/14 | 10626/15 | 10627/16 | 10628/17 | 10640/18 | 10629/19 | 10630/20 | 10631/21 | 10632/22 | 10633/23 | 10641/24 | 10635/25 | 10636/26 |
|------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| AM Peak          | E       | E       | E       | E       | E       | E       | E       | E       | E       | E        | E        | E        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        |
| AM Off Peak      | C       | D       | D       | E       | E       | E       | E       | E       | E       | E        | E        | E        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        |
| PM Peak          | D       | D       | D       | D       | E       | E       | E       | E       | E       | E        | E        | E        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        | F        |
| PM Off Peak      | D       | D       | D       | D       | D       | D       | D       | E       | E       | E        | E        | E        | E        | E        | E        | E        | E        | E        | F        | F        | F        | F        | F        | F        | F        | F        |

In the Inbound direction, the picture was broadly similar but with some variation by period. During the AM peak,  $CV_h$  values were already high near the start of the route and entered LOS F by approximately Stop 8, remaining at that level for most of the remaining corridor. The PM peak displayed the same broad instability, with reliability collapsing early and  $CV_h$  exceeding 1.0 in the later section. The AM off-peak and PM off-peak began from comparatively better conditions, with  $CV_h$  in the B-to-D range at the initial stops, but those gains were not sustained — both periods deteriorated steadily and reached LOS F in the latter half of the route. As in

Outbound, the all-day Inbound pattern shows a strong downstream loss of spacing control, with the worst reliability concentrated past the mid-route segment.

*Line Chart 8. Coefficient of variation of headways by stop for the Inbound direction across the four time periods.*

### AM Peak, AM Off Peak, PM Peak and PM Off Peak



*Table 6. Stop-level headway reliability LOS grades for the Inbound direction across the four time periods.*

| Stop sequence/ID | 106 43/1 | 106 35/2 | 106 34/3 | 106 33/4 | 106 32/5 | 106 31/6 | 106 30/7 | 106 29/8 | 106 28/9 | 106 27/10 | 106 26/11 | 106 25/12 | 106 24/13 | 106 23/14 | 106 22/15 | 106 21/16 | 106 20/17 | 106 19/18 | 106 18/19 | 106 17/20 | 106 15/21 | 106 16/22 | 106 14/23 | 106 13/24 | 106 42/25 |
|------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| AM Peak          | E        | D        | E        | E        | E        | E        | E        | F        | F        | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         |
| AM Off Peak      | B        | B        | C        | C        | D        | D        | D        | E        | E        | E         | E         | E         | E         | E         | E         | E         | E         | E         | E         | E         | F         | F         | F         | F         | F         |
| PM Peak          | D        | C        | D        | D        | E        | E        | E        | F        | F        | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         |
| PM Off Peak      | C        | C        | C        | D        | D        | D        | D        | D        | E        | E         | E         | E         | E         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         | F         |

Comparing the two directions, the headway-reliability problem on Route 99 is systemic rather than incidental. Buses are supplied at high frequency in both directions, yet spacing becomes

increasingly irregular as they progress along the corridor. Outbound exhibits a particularly strong downstream escalation during the PM peak; Inbound shows earlier deterioration in peak periods and broader unreliability across more of the day. The route is not failing because too few buses are deployed — it is failing because the buses that are deployed are not being held to regular intervals in operation. Even when average frequency remains short, poor spacing reduces predictability, increases waiting-time risk, and creates inconsistent crowding across the corridor.

#### **4.2.3 Expected Passenger Waiting Time ( $E[W]$ )**

Expected passenger waiting time translates headway irregularity into a direct passenger-facing measure. Under perfectly regular service, expected waiting time approaches one-half of the average headway; as variability rises, expected waiting time rises disproportionately, so passengers experience a poorer service than average frequency alone would suggest. In this section,  $E[W]$  is used to assess how the  $CV_h$  patterns above translate into actual waiting conditions along the corridor.

In the Outbound direction,  $E[W]$  began at favorable levels and deteriorated progressively toward the downstream end. During the AM peak, the first nine stops maintained LOS A; the route then shifted into LOS B and eventually LOS C at the terminus, where  $E[W]$  reached 4.48 minutes. The AM off-peak performed slightly worse overall, beginning at LOS B and transitioning to LOS C from the middle of the route onward. The PM peak showed the most severe deterioration, with  $E[W]$  rising from LOS A/B near the origin to LOS D and eventually LOS E in the final stops, reaching 6.95 minutes at the terminus. The PM off-peak also weakened, beginning at LOS C, dropping into LOS D in the latter section, and ending at LOS E. The all-period pattern confirms a clear downstream decline: LOS B in the first segment, LOS C through most of the route, and LOS D at the final stop.

Line Chart 9. Expected passenger waiting time by stop for the Outbound direction across the four time periods.

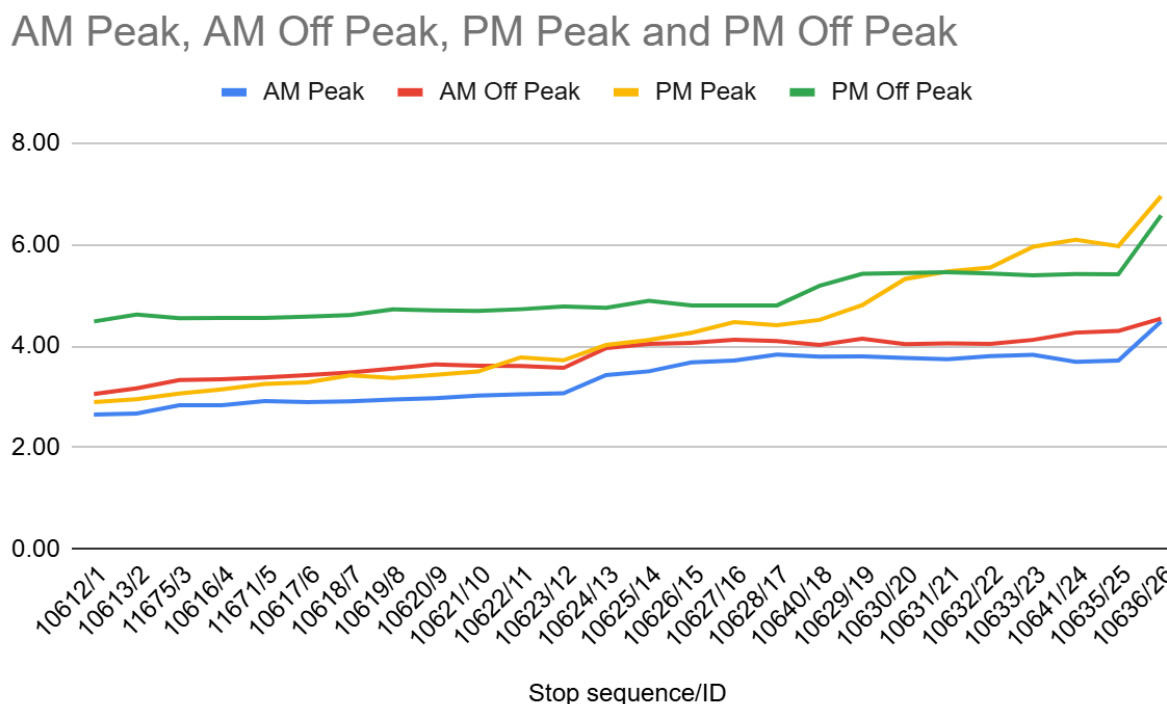


Table 7. Stop-level passenger waiting time LOS grades for the Outbound direction across the four time periods.

| Stop<br>sequence/ID | 106<br>12/<br>1 | 106<br>13/<br>2 | 116<br>75/<br>3 | 106<br>16/<br>4 | 116<br>71/<br>5 | 106<br>17/<br>6 | 106<br>18/<br>7 | 106<br>19/<br>8 | 106<br>20/<br>9 | 106<br>21/<br>10 | 106<br>22/<br>11 | 106<br>23/<br>12 | 106<br>24/<br>13 | 106<br>25/<br>14 | 106<br>26/<br>15 | 106<br>27/<br>16 | 106<br>28/<br>17 | 106<br>40/<br>18 | 106<br>29/<br>19 | 106<br>30/<br>20 | 106<br>31/<br>21 | 106<br>32/<br>22 | 106<br>33/<br>23 | 106<br>41/<br>24 | 106<br>35/<br>25 | 106<br>36/<br>26 |
|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| AM<br>Peak          | A               | A               | A               | A               | A               | A               | A               | A               | A               | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | B                | C                |
| AM Off<br>Peak      | B               | B               | B               | B               | B               | B               | B               | B               | B               | B                | B                | B                | C                | C                | C                | C                | C                | C                | C                | C                | C                | C                | C                | C                | C                | C                |
| PM<br>Peak          | A               | A               | B               | B               | B               | B               | B               | B               | B               | B                | B                | C                | C                | C                | C                | C                | C                | C                | C                | D                | D                | D                | D                | E                | D                | E                |
| PM Off<br>Peak      | C               | C               | C               | C               | C               | C               | C               | C               | C               | C                | C                | C                | C                | C                | C                | C                | C                | D                | D                | D                | D                | D                | D                | D                | D                | E                |

In the Inbound direction, the same overall pattern of downstream deterioration appeared, but severity varied more by period. The AM peak began at LOS A in the first three stops, then moved into LOS B and subsequently LOS C through much of the corridor. The AM off-peak was the most stable period in this direction, with LOS B across most of the route and only a late transition into LOS C toward the final stops. The PM peak declined more sharply, beginning at LOS A/B and deteriorating into LOS D in the latter third of the route. The worst service quality in this direction, however, occurred during the PM off-peak. In that period, the route started at LOS

B, shifted into LOS C, dropped to LOS D by approximately Stop 10, and reached LOS E in the final segment, where  $E[W]$  exceeded 6 minutes. The all-period pattern shows LOS B in the first eight stops, LOS C across most of the corridor, and LOS D at the end.

*Line Chart 10. Expected passenger waiting time by stop for the Inbound direction across the four time periods.*

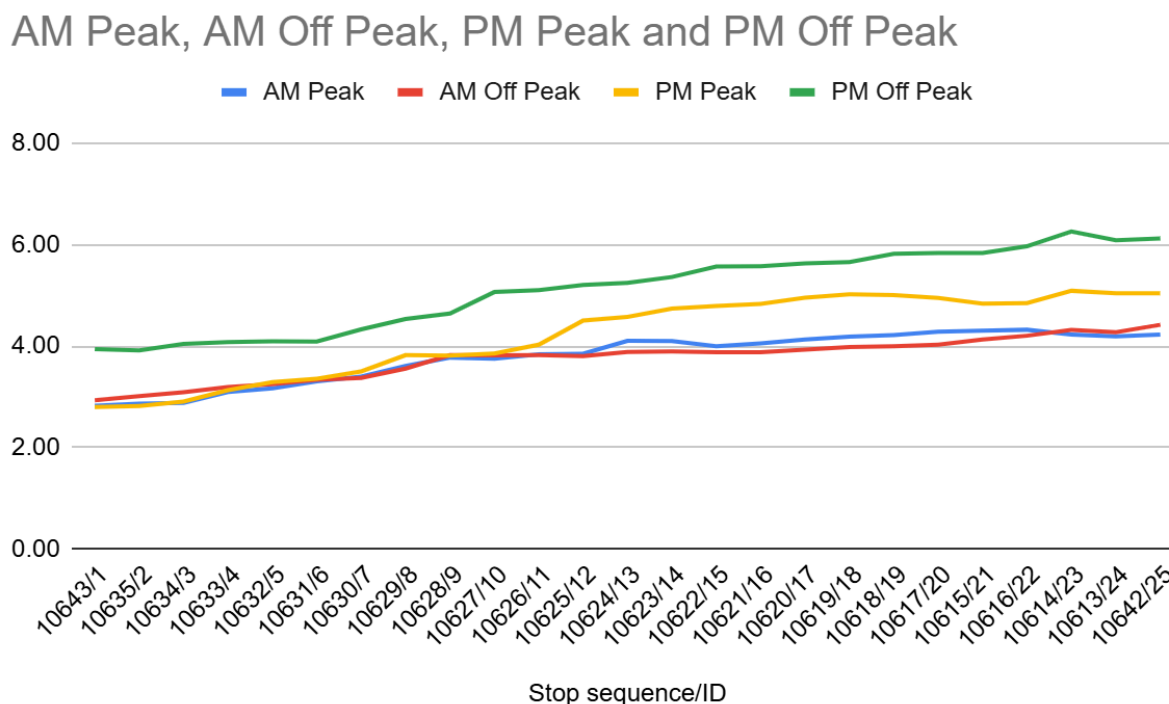


Table 8. Stop-level passenger waiting time LOS grades for the Inbound direction across the four time periods.

[illegible]

Across both directions,  $E[W]$  captures the practical consequence of unstable spacing more directly than mean headway does. Passengers near the origin terminals experience relatively favorable waiting conditions in most periods, but service quality weakens progressively as buses move downstream. Outbound exhibits its sharpest failure during the PM peak, when long and unreliable waits accumulate toward the terminus; Inbound performs worst during the PM off-peak, when lower frequency combines with persistent irregularity to produce poor waiting conditions in the latter half of the route.

#### 4.2.4 Synthesis

Across the three indicators, the headway results describe a fundamental contradiction in how Route 99 operates. Mean headways are short and uniformly graded LOS A throughout the day in both directions, so the service is in nominal terms, frequent enough to support a walk-up experience. That frequency advantage, however, is progressively undermined as buses move downstream.  $CV_h$  rises steadily along the corridor and reaches LOS F across large portions of the route in every period, and  $E[W]$  degrades in step with it, falling to LOS D or worse at the downstream end during the most affected periods in each direction. The corridor is providing frequent service but not consistently regular service.

The  $CV_h$  and  $E[W]$  patterns also show direction-specific failure modes. In Outbound, the most severe breakdown occurs during the PM peak, when downstream  $CV_h$  reaches extreme values and  $E[W]$  climbs to LOS D/E near the terminus. In Inbound, the same general instability is present, but the worst passenger-facing outcome appears during the PM off-peak, when reduced nominal frequency combines with persistent irregularity to produce long waits in the final segment. The underlying issue is similar in both directions — spacing control is not being maintained over the length of the route — but the worst time of day differs because the contributing conditions differ.

From a passenger perspective, the consequence is that the experience of using Route 99 becomes less predictable and less equitable as the trip progresses. Irregular spacing inflates expected waiting time, produces bunching and gapping, and distributes passengers unevenly across vehicles, so some buses become overloaded while others run lighter. The practical value of high frequency is therefore not fully realized, particularly past the mid-route segment in either direction. The route's central operational weakness on this dimension is not vehicle supply but spacing control over the corridor.

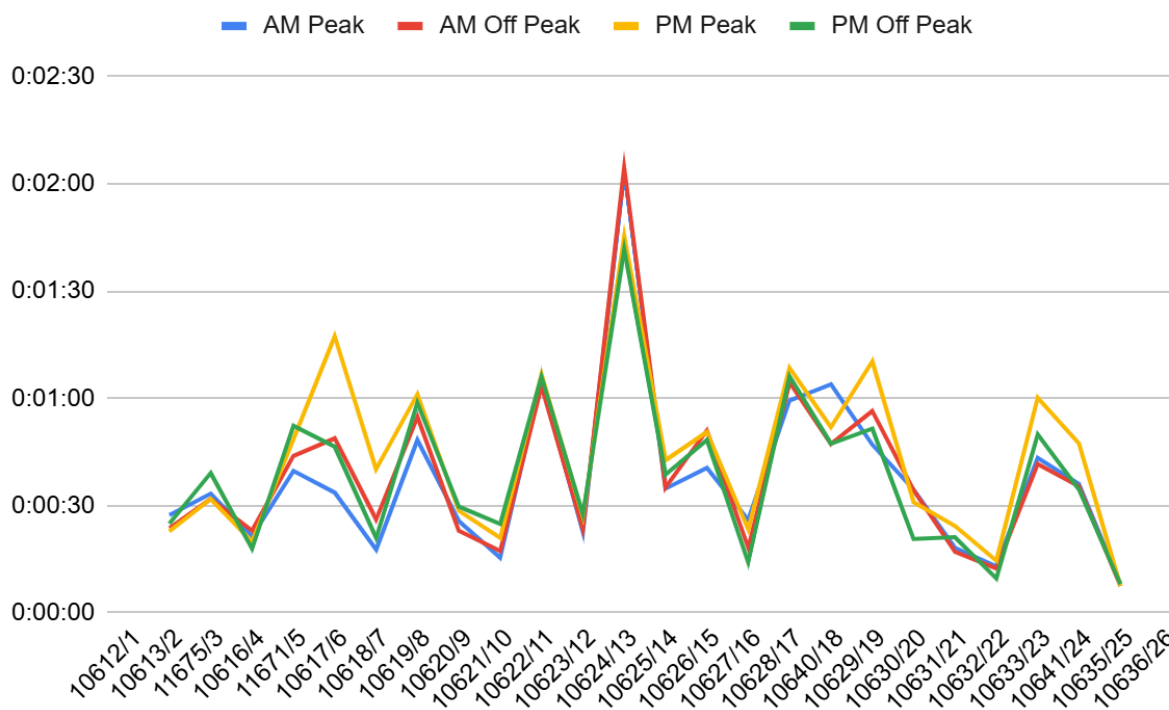
### 4.3 Dwell Time Analysis

Dwell time was analyzed to assess the contribution of stop-level operations to overall corridor reliability. In this study, dwell time refers to the period a bus remains stopped at a station while passenger boarding activity and associated operational processes take place. In high-frequency service, dwell time matters not only because long stops reduce operating efficiency, but also because inconsistent stop durations disrupt vehicle spacing and propagate headway instability downstream. The analysis combines four linked measures: average dwell time, the coefficient of variation of dwell time, average dwell time with and without recorded boardings, and the coefficient of determination ( $R^2$ ) between boarding activity and dwell time. Together, these are used to distinguish demand-driven stop delay from operational friction that cannot be attributed to passenger activity. Dwell-time values in this section are reported in seconds for consistency.

#### 4.3.1 Dwell-time patterns in the Outbound direction

The Outbound direction is characterized by a small number of persistent stop-level bottlenecks alongside generally moderate dwell performance elsewhere. The most prominent bottleneck is Stop 13 (Sport City Terminal), which recorded the highest average dwell in every observed period, ranging from 102 to 124 seconds across the four periods, with the highest values in the morning. The consistency of this pattern indicates that Sport City Terminal is not a context-specific outlier but a structurally constrained stop where service-processing demands remain persistently higher than elsewhere on the corridor.

*Line Chart 11. Average dwell time by stop for the Outbound direction across the four time periods.*



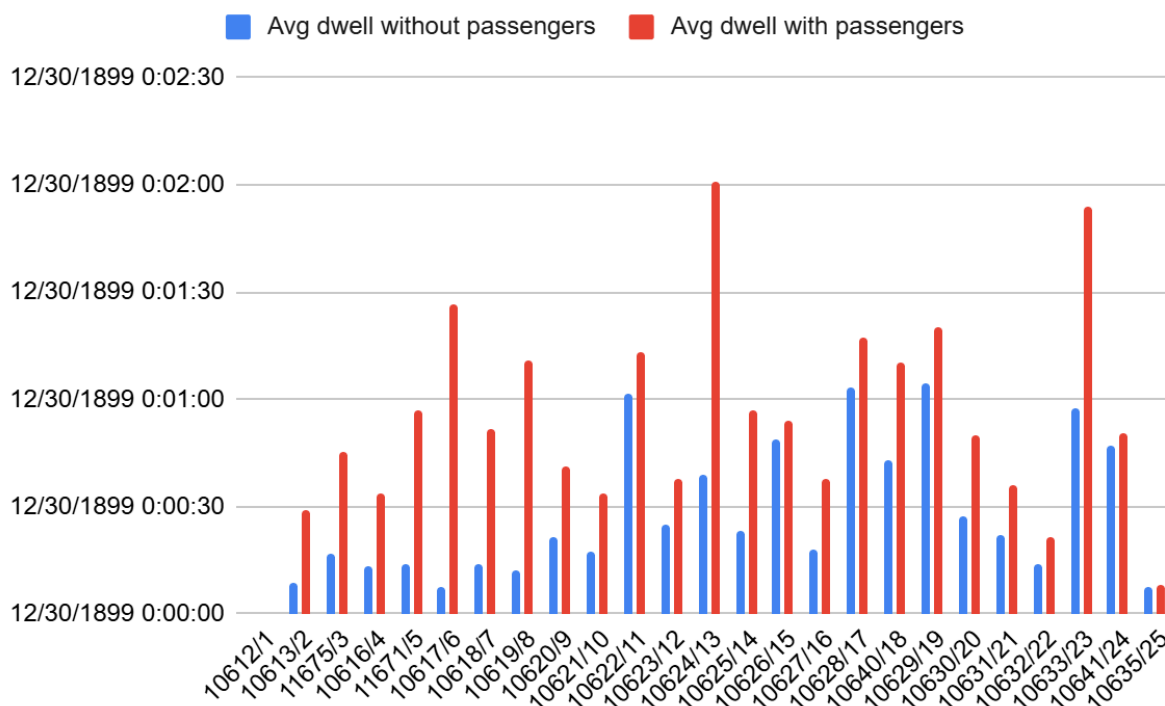
A secondary tier of elevated dwell-time locations recurs across multiple periods. Stop 11 (Alsahafeh Tunnel) averaged 63–67 seconds across all four periods, making it the most consistently elevated stop outside Sport City Terminal. Stop 17 (King Abdullah Gardens) followed a similar profile of 59–68 seconds across the day, peaking in the PM peak. Stop 18 (Wadi Saqra Int.) showed a different pattern — 64 seconds in the AM peak but closer to 50 in the other periods — suggesting its stop burden is more sensitive to the AM peak environment. Stop 19 (Mecca St.) displayed the opposite tendency, rising from 47 seconds in the AM peak to 70 in the PM peak. The corridor is therefore affected not by one isolated bottleneck but by a cluster of recurring friction points whose significance varies by time of day.

Stop-time variability adds another layer. Coefficient-of-variation values exceed 0.5 across many stops and frequently approach or exceed 0.7, peaking at 0.79 at Stop 22 in the AM peak and 0.94 at the same stop in the PM off-peak. These elevated CV values are not concentrated at the longest-dwelling stops; several appear at low-average-dwell locations, indicating that stop execution is highly inconsistent even where the burden is small. This is operationally important because such inconsistency injects variability into the headway sequence from the early stops onward, contributing to the downstream CVh growth documented in Section 4.2.

Separating dwell time by whether boarding activity was recorded provides further insight into the likely sources of stop delay. The ITS dataset records only boarding events, so dwell classified as "without passengers" reflects the absence of recorded boardings but may still include passenger alighting. With that caveat, the distinction is informative. At Sport City Terminal, with-boarding dwell ranged from 112 to 139 seconds across the four periods, while without-boarding dwell ranged from 19 to 52 — a substantial gap in every period that confirms a strong boarding-driven component, but with a non-trivial dwell remaining in the no-boarding condition that points to a residual operational floor at this stop.



*Bar Chart 12. Average dwell time with and without recorded boardings, PM peak, Outbound direction.*

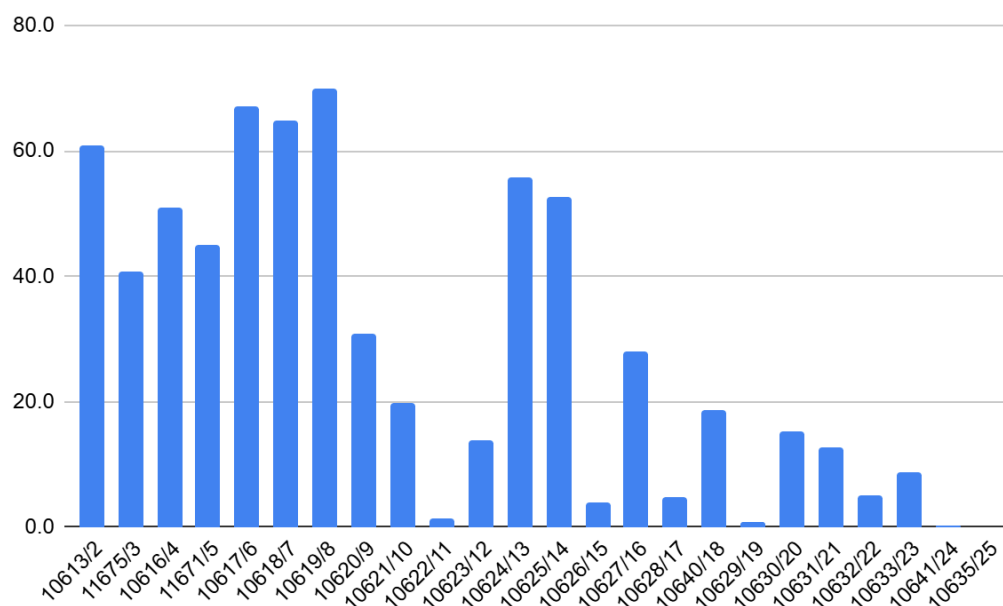


Boarding-driven effects are clearest at the university-adjacent stops in the first half of the corridor. During the PM peak, Stop 6 (University of Jordan) averaged 87 seconds with boardings against only 8 without, and Stop 8 (J.U. Hospital) averaged 71 against 12 — gaps large enough to leave little doubt that boarding activity is the dominant driver at these stops. By contrast, several downstream stops exhibited extended dwell even when no boarding was recorded. Stops 17, 18, and 19 all maintained substantial without-boarding dwells across the day, with Stop 19 reaching 64 seconds without recorded boardings in the PM peak. A substantial part of the stop burden in the latter half of the route therefore cannot be attributed to boarding and is more likely associated with operational friction, stop procedures, or local stop-environment conditions.

The  $R^2$  results reinforce this distinction. At Sport City Terminal,  $R^2$  ranged from 43.8% to 60.1% across the four periods, confirming that passenger demand explains a meaningful but incomplete share of dwell variation. Several first-half stops showed even stronger boarding-dwell relationships, with  $R^2$  values above 60% — including 79.6% at Stop 5 (Ahmad Altarawneh ST) in the AM off-peak and 70.0% at Stop 8 in the PM peak. By contrast, the downstream cluster showed near-zero values: Stop 17 recorded  $R^2$  below 10% in every period, Stop 18 stayed below 22%, and Stop 19 dropped to 1.1% in the AM peak and 0.8% in the PM peak. Stop 11 showed the same pattern (15.0% in the AM peak and only 1.3% in the PM peak) despite average dwells above one minute in both periods. The low explanatory values at these

high-dwell downstream locations strongly suggest that a meaningful share of the dwell burden in this direction is driven by operational factors rather than passenger boarding demand.

*Bar Chart 13. Correlation strength between passenger boardings and dwell time by stop during the PM peak in Direction 100990.*



The Outbound profile therefore describes a corridor in which both the magnitude and the source of stop delay vary by location. Sport City Terminal functions as a major recurring mixed-cause bottleneck where boarding activity explains a substantial share of dwell but does not fully account for the burden. The university-adjacent stops appear as demand-sensitive locations, especially during the PM peak. The King Abdullah Gardens–Wadi Saqra–Mecca Street cluster (Stops 17–19) exhibits extended dwell that is only weakly related to boardings, pointing to operational inefficiency rather than demand pressure as the dominant driver. A smaller number of stops with modest average dwell but high CV function less as bottlenecks than as instability injection points. Dwell-time reduction in this direction should therefore be differentiated by stop type: passenger-processing improvements at the boarding-intensive first-half stops, and procedural or operational interventions at the downstream cluster.

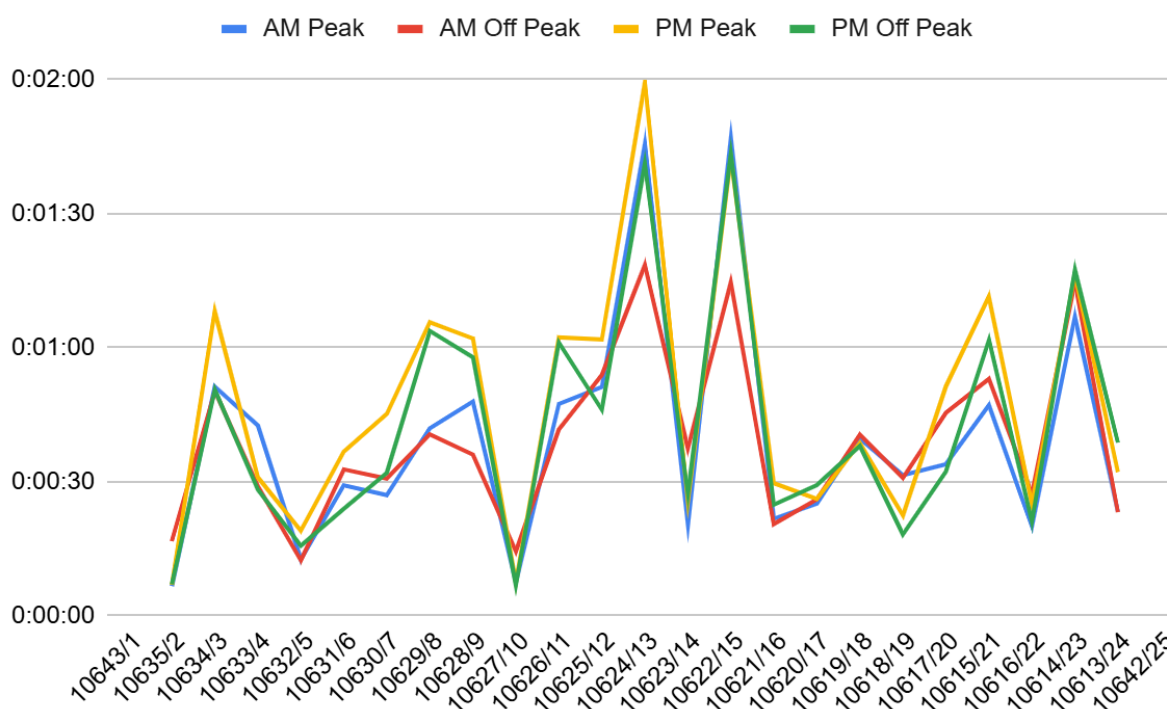
#### 4.3.2 Dwell-time patterns in the Inbound direction

The Inbound profile is more complex and operationally more severe. Rather than being dominated by a single bottleneck, this direction contains two persistently elevated stops of near-equal magnitude, together with a third recurring problem location. These impose a cumulative stop-level burden across the middle and later sections of the corridor and help explain the broader reliability problems observed in this direction in Section 4.2.

The two primary bottlenecks are Stop 15 (Alsaḥafeh Tunnel) and Stop 13. Stop 15 averaged 75–107 seconds across the four periods, with peak-period values close to 105 seconds. Stop 13

averaged 79–119 seconds across the same periods, with its PM peak value approaching two full minutes — the highest single-stop average in this direction. Where the Outbound direction was structured around one dominant bottleneck, this direction is constrained by a paired middle-route burden in which two consecutive stops each impose substantial and sustained delays.

*Line Chart 14. Average dwell time by stop for the Inbound direction across the four time periods.*

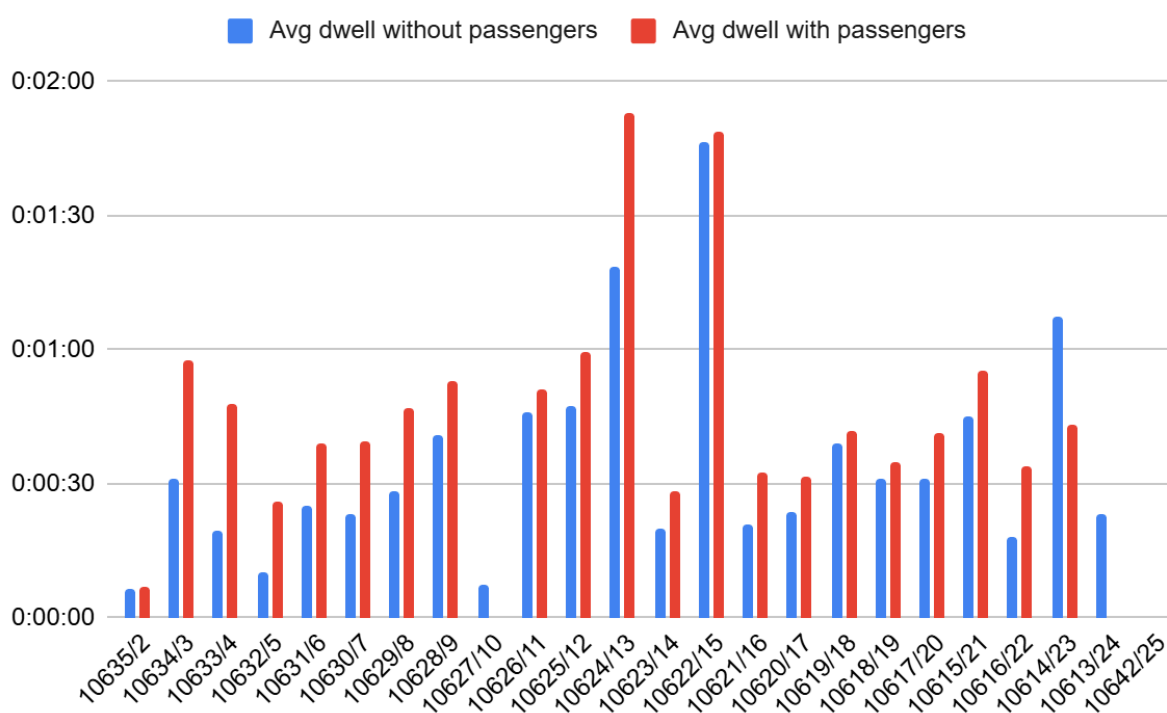


Stop 23 (Yajouz ST) forms a consistent secondary bottleneck, averaging 67–77 seconds across all four periods — a location-specific constraint that persists across the service day. Stop 3 (Prs. Basmah ST) is another notable location, averaging around 50 seconds outside the PM peak but rising to 68 seconds during it. Stop 21 (Ahmad Altarawneh ST) also becomes more burdensome in the afternoon, rising from around 50 seconds in the morning periods to 71 in the PM peak and 62 in the PM off-peak. The dwell burden in this direction is therefore distributed across several recurring stops rather than concentrated at a single location.

Stop-time variability in this direction is notably more extreme than Outbound. CV values exceeding 1.0 appear at multiple stops, particularly during the off-peak periods: 1.64 at Stop 2 and 1.41 at Stop 10 in the AM off-peak, and a peak of 1.98 at Stop 24 in the PM off-peak. A CV at or above 1.0 indicates that the standard deviation of dwell time equals or exceeds the mean — operationally, stop times are close to random. The concentration of extreme CV values in the off-peak periods is itself worth noting: it suggests that the consistency of stop execution deteriorates further when passenger loads are lighter, rather than improving with them.

The with- versus without-boarding comparison provides the clearest diagnostic distinction between the two primary bottlenecks. At Stop 15 the pattern is striking: in the AM peak, dwell without recorded boardings was 106 seconds and dwell with boardings was 109 seconds — a difference of only 3 seconds. The gap remained narrow across the other periods (under 20 seconds even in the PM peak, when boarding activity is highest), and even the AM off-peak without-boarding value remained at 69 seconds. The large dwell at Stop 15 is therefore only weakly affected by boarding activity and is more consistent with a process-driven stop delay.

*Bar Chart 15. Average dwell time with and without recorded boardings, AM peak, Inbound direction.*

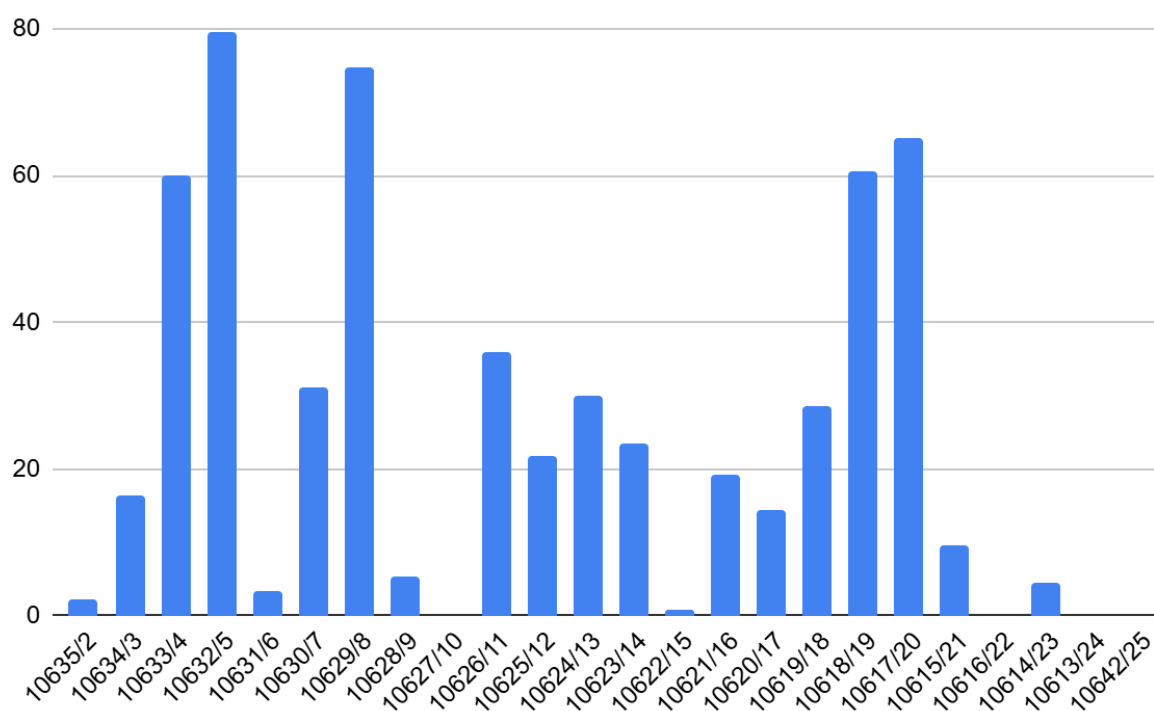


At Stop 13, the picture differs. Without-boarding dwell was also elevated, but the with-boarding values were consistently and substantially higher, ranging from 89 to 122 seconds across the four periods. In the PM peak, the gap between the two conditions reached 31 seconds and the with-boarding average exceeded two minutes. Stop 13 therefore combines a substantial operational floor with a meaningful boarding-related component — a mixed-cause rather than purely process-driven bottleneck. Stop 23 behaves more like Stop 15 than Stop 13, with its without-boarding dwell remaining between 67 and 77 seconds across the peak and off-peak periods, indicating persistent operational overhead at this location.

The  $R^2$  results reinforce these distinctions. At Stop 15,  $R^2$  stayed below 4% in every period, including 0.1% in the AM peak and 0.9% in the PM peak — values low enough that almost none of the dwell variation at this stop is explained by boarding count. Combined with the near-equality of dwell with and without boardings, this provides strong evidence that Stop 15's

burden is primarily operational rather than demand-driven. Stop 13 recorded  $R^2$  between 30.0% and 45.1% across the four periods, consistent with a mixed-cause profile. Stop 23 stayed below 5% in three of four periods (rising only to 20.5% in the AM off-peak), confirming that its dwell is only weakly related to boarding activity. By contrast, several stops showed strong and relatively stable boarding-dwell relationships: Stop 4 (Barda ST) recorded  $R^2$  between 59.3% and 73.2% across the four periods, making it the most consistently demand-sensitive stop in this direction, and Stop 8 also showed high values, especially in the PM periods.

*Bar Chart 16. Correlation strength ( $R^2$ ) between boardings and dwell time by stop, PM peak, Inbound direction.*



The Inbound profile therefore differs from Outbound in both its structure and its implications. Stops 15 and 23 behave as operationally driven bottlenecks — high average dwell, weak boarding-dwell relationships, substantial no-boarding floors. Stop 13 is mixed-cause. A wider set of stops carries moderate but recurring dwell burdens across the service day, and stop-time variability is more extreme, particularly off-peak. Improvements in this direction must therefore address both persistent bottlenecks and recurring process inefficiency: passenger-processing measures may suit the more demand-sensitive stops, but stronger operational control and more standardized stop procedures appear especially important here, where several high-dwell locations remain only weakly related to boarding activity.

#### 4.3.3 Synthesis

The dwell-time results show that stop-level delays on Route 99 are not uniform across the corridor and do not arise from a single type of operational problem. The two directions exhibit

different dwell-time structures, and those differences help explain the directional asymmetries observed earlier in schedule adherence and headway reliability.

The Outbound direction is anchored by one dominant recurring bottleneck at Sport City Terminal, supported by a downstream cluster of friction-dominated stops. Sport City Terminal is mixed-cause: boarding demand explains a meaningful share of its dwell, but a non-trivial operational floor remains. The downstream cluster (Stops 17–19) shows elevated dwell with weak boarding-dwell relationships, pointing to operational rather than demand-driven causes. The Inbound direction is more compounded: two paired middle-route bottlenecks (Stops 13 and 15), a third recurring problem location (Stop 23), and a wider set of moderate but recurring dwell burdens across the route. Among the primary Inbound bottlenecks, Stop 13 is mixed-cause while Stops 15 and 23 are most consistent with purely operational delay. This broader and more distributed burden helps explain why Inbound shows more persistent service degradation across the corridor rather than a decline centered mainly in the downstream segment.

A second distinction concerns variability. In Outbound, dwell-time inconsistency is concentrated at a smaller number of high-burden stops, with some downstream locations also displaying weak passenger linkage. In Inbound, the variability problem is broader: dwell-CV values exceeding 1.0 at multiple stops — including several with low average dwell — indicate that stop execution is highly inconsistent independent of the time being consumed. This is the bridge to the Section 4.2 findings on headway reliability. Inconsistent stop times propagate directly into spacing variability, and the dwell-CV pattern is consistent with the CVh growth observed along both corridors. The dwell-time results thus help account for headway irregularity that mean headway alone could not explain.

Overall, dwell-time interventions on Route 99 cannot be uniform across stops or directions. Boarding-intensive stops are more likely to benefit from passenger-processing improvements, while operationally driven bottlenecks require tighter procedural control and more standardized stop processes. Dwell time emerges in this analysis not only as a performance outcome but as a diagnostic lens through which the route's wider reliability failures can be understood.

## 4.4 Operational Efficiency and Passenger Comfort

Operational efficiency and passenger comfort were assessed to examine how Route 99 performs at the trip level under varying demand conditions. While the preceding sections focused on reliability, headway regularity, and stop-level delay, this section considers the broader service outcome in terms of passenger flow, trip time, crowding, and operating speed. Together, these indicators link internal operational behavior to the passenger experience of the route, showing how demand pressure and limited recovery margin affect both service efficiency and onboard comfort. Because the ITS dataset records boarding activity but not passenger alighting, the passenger flow and load factor measures presented in this section are interpreted at the trip level rather than as exact stop-by-stop onboard occupancy. These indicators are therefore used as proxies for overall demand intensity and vehicle utilization across the route. Trip times are reported in decimal minutes for ease of comparison across periods.

### 4.4.1 Operational efficiency and passenger comfort — Outbound direction

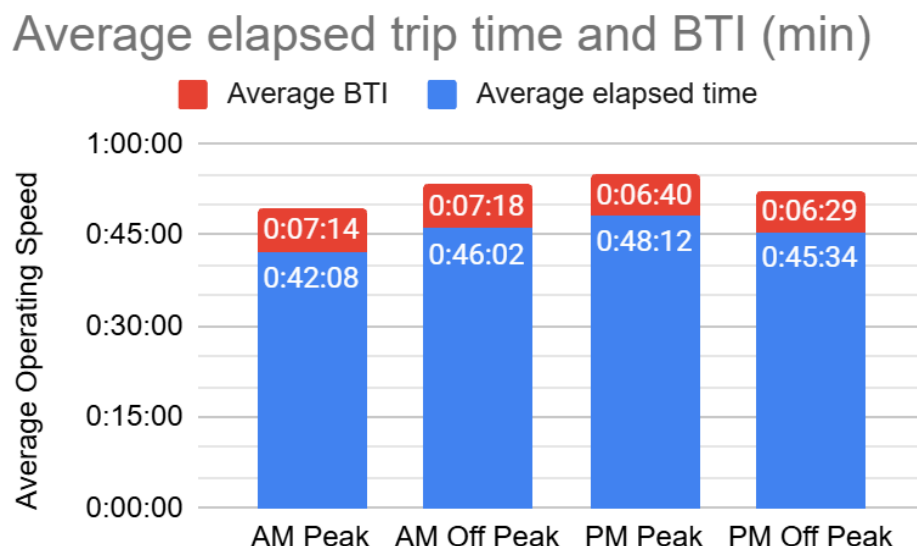
#### *Trip Time and Travel Time Reliability*

Average elapsed trip time increased from morning to afternoon. In the AM peak the mean trip duration was 42.1 minutes, rising to 46.0 in the AM off-peak, 48.2 in the PM peak, and then declining slightly to 45.6 in the PM off-peak. The AM off-peak average exceeded the PM off-peak average, which is noteworthy: midday conditions in this direction do not produce materially faster service than the early evening period.

Trip-time variability, expressed as the coefficient of variation of elapsed time, remained low across all four periods — between 0.11 and 0.14 — indicating that trip duration was relatively consistent from one run to the next within each period. The service does not produce highly scattered travel times; instead, trip durations cluster relatively tightly around the period average.

The Buffer Time Index (BTI) ranged from 0.14 to 0.17, highest in the AM peak (0.17) and lowest in both PM periods (0.14). As defined in the methodology, BTI represents the additional proportion of travel time a passenger would need to allow, relative to the mean trip time, to arrive on time with 95% probability. A BTI of 0.17 in the AM peak therefore implies that a passenger should allow approximately 17% more than the average trip time to travel reliably in that period. The consistently low BTI values across all four periods suggest that trip times are tightly clustered around the mean and that the route operates with relatively limited recovery margin. In practical terms, the service is not highly volatile in travel time, but it also leaves little tolerance for additional disruption once delays begin to accumulate.

*Line Chart 17. Average elapsed trip time and Buffer Time Index for the Outbound direction across the four time periods.*



### ***Load Factor and Passenger Comfort***

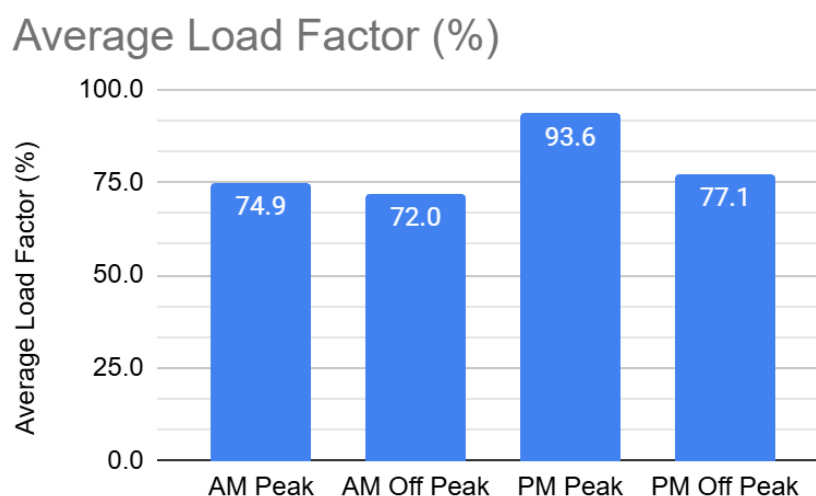
The average cumulative trip load factor, defined here as total boardings per trip as a percentage of vehicle capacity, produced LOS C in three of the four periods and LOS D in the PM peak. The AM peak (74.9%), AM off-peak (72.0%), and PM off-peak (77.1%) all sit within LOS C, indicating moderate trip-level utilization across most of the day.

The PM peak, by contrast, produced an average load factor of 93.6% (LOS D) — the only period in which the Outbound direction crossed into the lower comfort band. Average trip-level utilization during the afternoon peak therefore approached the practical capacity of the vehicle. With a standard deviation of 32.6 percentage points, the PM peak also suggests that a substantial share of trips likely operated at or above full capacity on a cumulative basis. In operational terms, the service relies on rapid passenger turnover to sustain loading without severe crowding on individual segments, and any further increase in demand or concentration of passengers due to irregular service would place additional pressure on comfort.

The coefficient of variation of load factor was 0.39 in the AM peak, 0.40 in both the AM off-peak and PM off-peak, and 0.30 in the PM peak. The lower CV during the PM peak, despite the highest absolute loading, suggests that peak demand is more uniformly distributed across trips. The higher CV in the off-peak periods indicates more uneven trip-to-trip loading, with some runs carrying substantially heavier demand than others — consistent with the headway irregularity documented in Section 4.2, where bunching and gapping would be expected to produce uneven passenger distribution across vehicles even when aggregate demand remains moderate.



*Line Chart 18. Average load factor (%) for the Outbound direction across the four time periods.*



### ***Operating Speed***

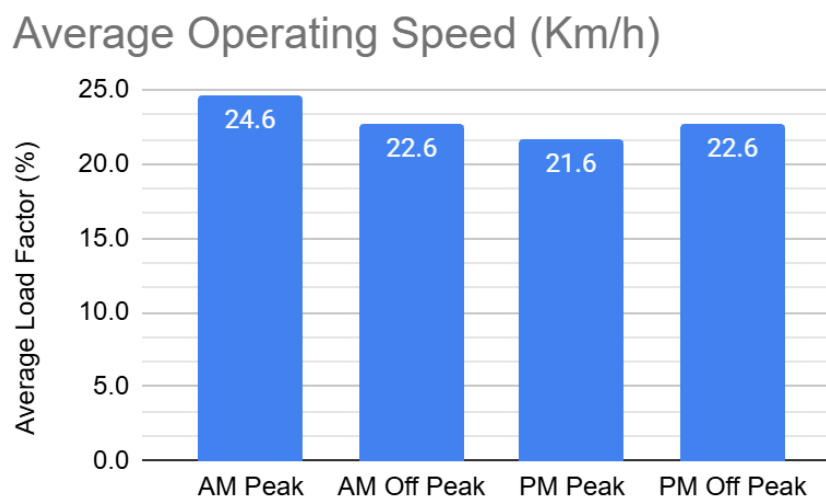
Operating speed declined progressively from the AM peak to the PM peak and recovered only partially in the PM off-peak. Mean operating speed was 24.6 km/h in the AM peak, 22.6 in the AM off-peak, 21.6 in the PM peak, and 22.6 in the PM off-peak. These values place the route within the applied LOS C to D range across the service day, indicating that it operates below the level expected of a highly competitive BRT corridor.

The AM peak was the fastest period despite its relatively high demand. This is consistent with the shorter average trip time in that period and suggests that the combined effect of morning operating conditions and lower aggregate stop burden still permits somewhat better performance than later in the day. By contrast, the PM peak produced both the highest dwell burden and the longest average elapsed trip time, making it the clear period of minimum operational efficiency.

The coefficient of variation of operating speed was 0.12 in the AM peak, 0.20 in both the AM off-peak and PM peak, and 0.10 in the PM off-peak. These relatively low values indicate that speed suppression is not occasional but a stable condition of service. The route is not intermittently slow; it is consistently slow, with the PM peak representing the clearest low-speed condition.

The relationship between load factor and operating speed across the four periods shows a broad negative association: higher average demand coincides with lower average speed. The AM peak and PM peak anchor the two ends of this pattern, while the off-peak periods sit in between. However, the AM off-peak and PM off-peak both record the same average speed of 22.6 km/h despite different average load factors, indicating that passenger volume alone does not fully explain speed suppression. Other factors, including traffic conditions and the dwell-time patterns identified in Section 4.3.1, also contribute materially to reduced speed in this direction.

*Line Chart 19. Average operating speed for the Outbound direction across the four time periods.*



### **Synthesis**

The Outbound efficiency and comfort results describe a route that maintains adequate average performance in most periods but experiences a concentrated deterioration in the PM peak. This period simultaneously produced the highest demand, the slowest operating speed, the longest trip time, and the only LOS D load factor in the direction, confirming that it places the route under its greatest operational strain. At the same time, the low elapsed-time CVs and BTI values across all periods indicate that the service produces relatively stable trip durations. The problem is therefore not that the route is unpredictably slow, but that it is consistently slow and consistently heavily loaded under peak demand.

The persistent speed suppression across all periods, between 21.6 and 24.6 km/h, reflects the combined effect of traffic conditions and the stop-level dwell burden identified earlier in Section 4.3.1. Reducing dwell time at the principal bottleneck stops, particularly through measures that reduce the boarding-time burden, represents the most direct controllable lever for improving operating speed without altering the broader operating environment. The PM peak load factor of 93.6% also indicates that capacity management during that window remains necessary if passenger comfort is to be protected from further deterioration under either demand growth or increased service irregularity.

### **4.4.2 Operational efficiency and passenger comfort — Inbound direction**

#### ***Trip Time and Travel Time Reliability***

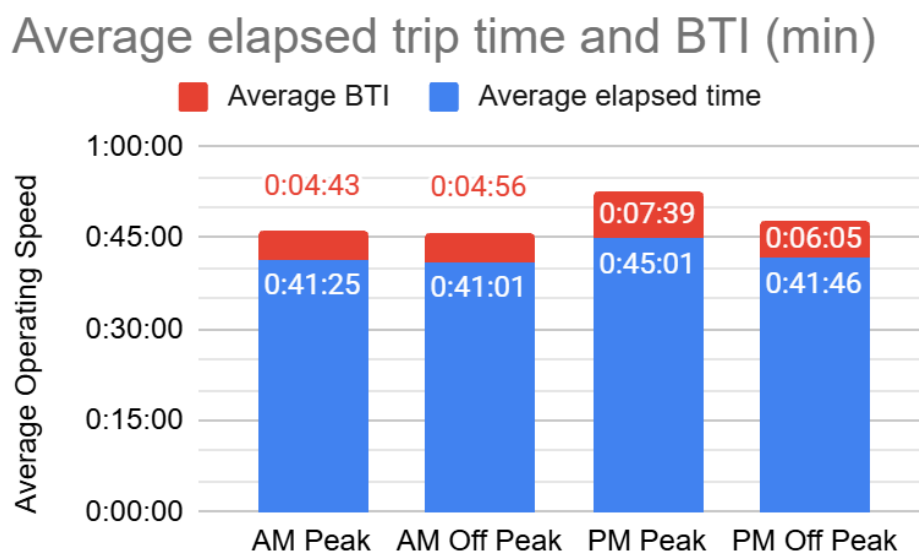
The trip-time profile of the Inbound direction is defined by a stable baseline in three periods and a clear deterioration during the PM peak. Average elapsed trip times were 41.4 minutes in the AM peak, 41.0 in the AM off-peak, 45.0 in the PM peak, and 41.8 in the PM off-peak. Three of the four periods therefore fall within a narrow range of less than one minute, while the PM peak extends journey time by approximately four minutes relative to that baseline. Unlike Outbound,

Inbound does not experience a gradual intraday lengthening of travel time; its trip-time deterioration is concentrated in a single period.

Trip-time variability was low across all four periods, with the coefficient of variation of elapsed time ranging from 0.08 in the AM off-peak to 0.12 in the PM peak. The PM peak therefore showed the highest variability as well as the longest mean trip time, suggesting that the afternoon deterioration is not only slower on average but also slightly less internally stable than the three other periods.

The Buffer Time Index reinforces this pattern. Values were 0.11 in the AM peak, 0.12 in the AM off-peak, 0.17 in the PM peak, and 0.15 in the PM off-peak. The AM peak value of 0.11 is the lowest recorded across all periods in both directions, indicating that morning trip times in Inbound were especially concentrated around their mean. The PM peak value of 0.17 is the highest for this direction and reflects the wider spread introduced by afternoon congestion and stop-level delay accumulation. The pattern indicates that observed trip durations in Inbound are tightly clustered outside the PM peak and become more dispersed once afternoon operating pressure increases.

*Line Chart 20. Average elapsed trip time and Buffer Time Index for the Inbound direction across the four time periods.*



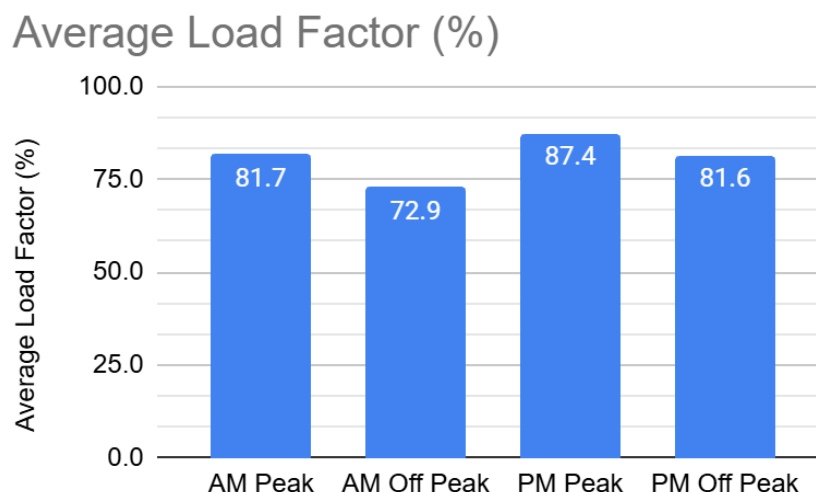
### ***Load Factor and Passenger Comfort***

Passenger demand in Inbound was consistently high throughout the day and more evenly distributed across periods than in Outbound. Average boardings per trip ranged from 42.6 (AM off-peak) to 49.7 (PM peak), with the AM peak (48.0) and PM off-peak (47.8) close to the upper end. Three of the four periods therefore clustered tightly close to 48 boardings, with only the AM off-peak showing a noticeable reduction. Inbound operates under sustained demand pressure across most of the day rather than experiencing a sharply isolated peak.

This demand structure is reflected in the load factor results. Average cumulative trip load factor was 81.7% in the AM peak, 72.9% in the AM off-peak, 87.4% in the PM peak, and 81.6% in the PM off-peak. Under the applied LOS interpretation used in this study, this corresponds to LOS C in three periods and LOS D in the PM peak. The PM peak therefore remains the primary comfort failure period in this direction, but the high values in the AM peak and PM off-peak indicate that passenger loading remains elevated outside the peak as well. The near-equality between AM peak and PM off-peak load factors is especially important: it suggests that Inbound operates close to its comfort threshold for much of the service day rather than only during the afternoon peak.

The coefficient of variation of load factor was 0.30 in all four periods. This uniformity indicates that the degree of trip-to-trip loading variability was essentially unchanged across the service day. In practical terms, the risk of heavily loaded individual trips is not concentrated only in the PM peak — even where average utilization remains within LOS C, the route still experiences a persistent risk of uneven crowding across trips. This is consistent with the headway irregularity documented earlier in Section 4.2.

*Line Chart 21. Average load factor (%) for the Inbound direction across the four time periods.*



### ***Operating Speed***

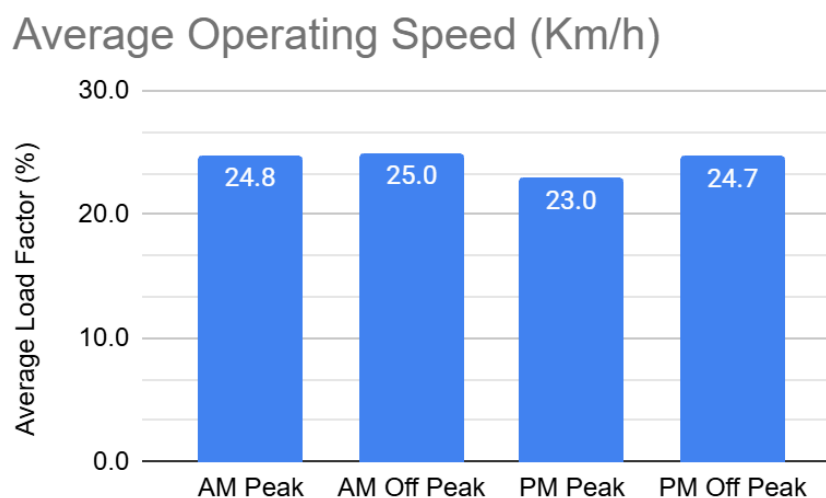
Operating speed remained slightly higher in Inbound than in Outbound across all four periods, but it still indicates a service operating below the standard of a high-performance BRT corridor. Average speed was 24.8 km/h in the AM peak, 25.0 in the AM off-peak, 23.0 in the PM peak, and 24.7 in the PM off-peak. All four values remain within the applied LOS C range, unlike Outbound, where the PM peak dropped into LOS D.

The PM peak again emerges as the lowest-efficiency period, with the slowest operating speed and the longest elapsed trip time. This is consistent with the dwell-time findings in Section 4.3.2, where the direction exhibited multiple bottlenecks in the middle and later portions of the route and where stop-level burdens intensified under PM conditions. The AM off-peak produced the

fastest average speed of the day at 25.0 km/h, consistent with its shorter elapsed time and lower average demand. The PM off-peak speed of 24.7 km/h was nearly identical to the AM peak value, reinforcing the broader finding that three of the four periods operate within a relatively narrow efficiency range while the PM peak stands apart as the sole low-speed deterioration event.

The coefficient of variation of operating speed was 0.10 in all four periods. This contrasts with Outbound, where speed variability differed more meaningfully by period. The consistency of the Inbound speed CV indicates that the spread of operating speeds around the mean remained stable across the day — speed is not only moderately low but also predictably so, suggesting that the route is shaped by relatively constant operational constraints rather than highly volatile conditions.

*Line Chart 22. Average operating speed for the Inbound direction across the four time periods.*



## Synthesis

The Inbound operational profile is defined by two main characteristics. First, trip-time performance outside the PM peak is remarkably stable, with the AM peak, AM off-peak, and PM off-peak all operating within a narrow travel-time range and with low BTI values. Second, the PM peak represents a clear deterioration event rather than part of a gradual intraday decline: trip times lengthen, BTI rises, load factor crosses into LOS D, and operating speed reaches its minimum, all within the 14:00 to 18:00 period.

Passenger loading introduces a broader concern. Unlike Outbound, where the PM peak stood further apart from the rest of the day, Inbound remains highly loaded across most periods. The AM peak and PM off-peak both operate at approximately 81.6% to 81.7% load factor, and the load factor CV remains constant at 0.30 throughout the day. The route is therefore operating close to its comfort threshold not only in the peak but across much of the service day. Read alongside the dwell-time findings from Section 4.3.2, the picture is of a direction where sustained demand interacts with recurring stop-level inefficiency to produce a corridor that

remains close to its performance limit even outside the PM peak. The PM peak is the clearest failure period, but the underlying pressure that produces it is present throughout the day.

#### 4.4.3 Synthesis

The operational efficiency and passenger comfort results show that both directions of Route 99 operate under consistently high demand, but they do so in different ways and with different types of performance pressure. Outbound is characterized by a more pronounced PM peak deterioration, in which demand, trip time, load factor, and operating speed all worsen together. Inbound, by contrast, operates under a more sustained demand burden across the day, with less separation between peak and non-peak loading conditions and a more stable trip-time baseline outside the PM peak. In both directions, the PM peak remains the clearest period of performance stress, but the nature of that stress differs: in Outbound it appears as a sharper peak-period compression of efficiency and comfort, whereas in Inbound it emerges from a corridor that is already operating close to its loading threshold in multiple periods.

The trip-time results reinforce this distinction. Outbound shows a broader intraday spread in elapsed time, rising from 42.1 minutes in the AM peak to 48.2 in the PM peak, before declining only partially in the PM off-peak. Inbound maintains a relatively stable trip-time baseline in the AM peak, AM off-peak, and PM off-peak, with the PM peak standing out as a discrete deterioration event. Outbound therefore experiences a more gradual degradation in trip efficiency across the day, while Inbound is more stable outside the PM peak but still experiences a marked breakdown once afternoon pressure intensifies. The Buffer Time Index values support this interpretation: both directions operate with relatively low BTI overall, but Inbound records the lowest BTI in the study during the AM peak, while Outbound remains slightly more exposed to trip-time spread outside the PM peak.

Passenger comfort results reveal a parallel directional contrast. Both directions cross into LOS D during the PM peak, but Outbound does so at a more severe average load factor of 93.6%, compared with 87.4% in Inbound. Outbound is therefore more acutely overloaded during the afternoon peak and more vulnerable to crowding-related service degradation in that window. Inbound, however, shows a different form of capacity pressure: high load factors persist outside the PM peak, with the AM peak and PM off-peak both remaining at approximately 81.6% to 81.7%. Outbound shows the stronger peak-period crowding failure, whereas Inbound shows a more sustained utilization burden across the service day.

Operating speed produces a similar comparison. Inbound remains marginally faster than Outbound in every period, but both directions operate within a relatively constrained speed range and both record their lowest speeds in the PM peak. The difference is that Outbound drops to 21.6 km/h in the PM peak, producing the corridor's clearest low-speed condition, while Inbound remains slightly higher at 23.0 km/h. The speed penalty imposed by dwell burden, traffic conditions, and operational friction is therefore somewhat more severe in Outbound. The low coefficients of variation for operating speed in both directions indicate that this is not an occasional or isolated problem: the route is not unpredictably slow, but consistently slow, particularly under peak demand.

Read alongside the dwell-time analysis in Section 4.3, these efficiency results become more interpretable. In Outbound, the PM peak failure appears closely tied to the combination of one

dominant mixed bottleneck, several downstream friction points, and the highest passenger loading of the day. In Inbound, the PM peak deterioration is still evident, but it occurs within a direction that already carries high passenger demand and recurring stop-level inefficiency outside the peak. The operational envelope of Inbound is therefore persistently constrained, while Outbound is more clearly pushed into failure by the concentrated PM peak load.

Overall, the synthesis of operational efficiency and passenger comfort confirms that the main challenge on Route 99 is not underuse of service but operation close to capacity under constrained speed conditions and limited recovery margin. The two directions do not fail in identical ways. Outbound shows a sharper and more concentrated PM peak stress pattern, especially in load factor and speed; Inbound shows a more persistent all-day utilization burden with a distinct PM peak deterioration layered on top of it. The implication is that interventions should also differ by direction: Outbound requires stronger PM peak capacity and speed recovery measures, while Inbound requires both PM peak management and broader attention to sustained loading and recurring stop-level inefficiency across the day.



## 4.6 Composite LOS Index

The preceding sections of this chapter evaluated the operational performance of Route 99 across several distinct dimensions, including reliability, service frequency, dwell-time behavior, operating speed, and passenger comfort. While these analyses identified specific strengths and failure modes, they do not by themselves indicate how the route performs as an overall service. The composite Level of Service (LOS) index addresses this by aggregating the principal dimensions into a single weighted score, allowing the overall quality of each direction to be expressed in a form that supports direct comparison and summary assessment.

### 4.6.1 LOS Grade Numerical Conversion

To enable arithmetic aggregation, the alphabetical LOS grades were converted to a numerical scale in which A corresponds to 5 and F to 0, with equal unit intervals between grades. This conversion preserves the ordinal ranking of the original categories while allowing weighted summation across components. Final composite scores were then mapped back to letter grades using equal-interval boundaries on the 0 to 5 scale, with LOS C defined as the range from 2.50 to 3.49.

*table 22. LOS Grade in Numerical Value.*

| LOS Grade       | A | B | C | D | E | F |
|-----------------|---|---|---|---|---|---|
| Numerical Value | 5 | 4 | 3 | 2 | 1 | 0 |

### 4.6.2 Component LOS Scores

Four components contribute to the composite index: reliability (30%), frequency (30%), operating speed (20%), and passenger comfort (20%). Reliability and frequency were given equal weight because they jointly shape the passenger experience at the point of access, with one determining how predictably the vehicle arrives and the other determining how frequently it is available. Speed and comfort were weighted equally at 20%, reflecting their importance to the in-vehicle experience while recognizing that the most immediate operational failures on Route 99 are concentrated in reliability and service regularity. These weights are the author's own applied specification for this study. The TCQSM does not prescribe a composite weighting method, so the index presented here should be interpreted as an applied evaluation tool rather than a direct TCQSM output.

For Direction 100990 (Swaileh to Jordan Museum), the reliability component was derived from the stop-level headway reliability grades reported in Section 4.3. Because most stops were dominated by LOS F conditions, the average numerical reliability score was 1.00, corresponding to LOS E. Frequency, by contrast, scored a perfect 5.00, since mean observed headway

remained LOS A across all stops and periods. The speed component averaged 2.75, reflecting LOS grades of C, C, D, and C across the four time periods, while the comfort component also averaged 2.75, based on load factor grades of C, C, D, and C.

For Direction 100991 (Jordan Museum to Swaileh), the reliability component was slightly higher at 1.60, though it still mapped to LOS E overall. This reflects the fact that some early stops retained somewhat better headway control in selected periods before deteriorating further downstream. Frequency again scored 5.00, since LOS A prevailed across all stops and periods. The speed component averaged 3.00, reflecting LOS C in all four periods, while the comfort component matched Direction 100990 at 2.75, again corresponding to C, C, D, and C across the day.

#### 4.6.3 Composite Index and Final Grades

Applying the weighted formula

$$LOS_{\text{overall}} = (0.30 \times R) + (0.30 \times F) + (0.20 \times S) + (0.20 \times C)$$

produced the following results:

##### For Direction 100990:

$$(0.30 \times 1.00) + (0.30 \times 5.00) + (0.20 \times 2.75) + (0.20 \times 2.75) = 2.90$$

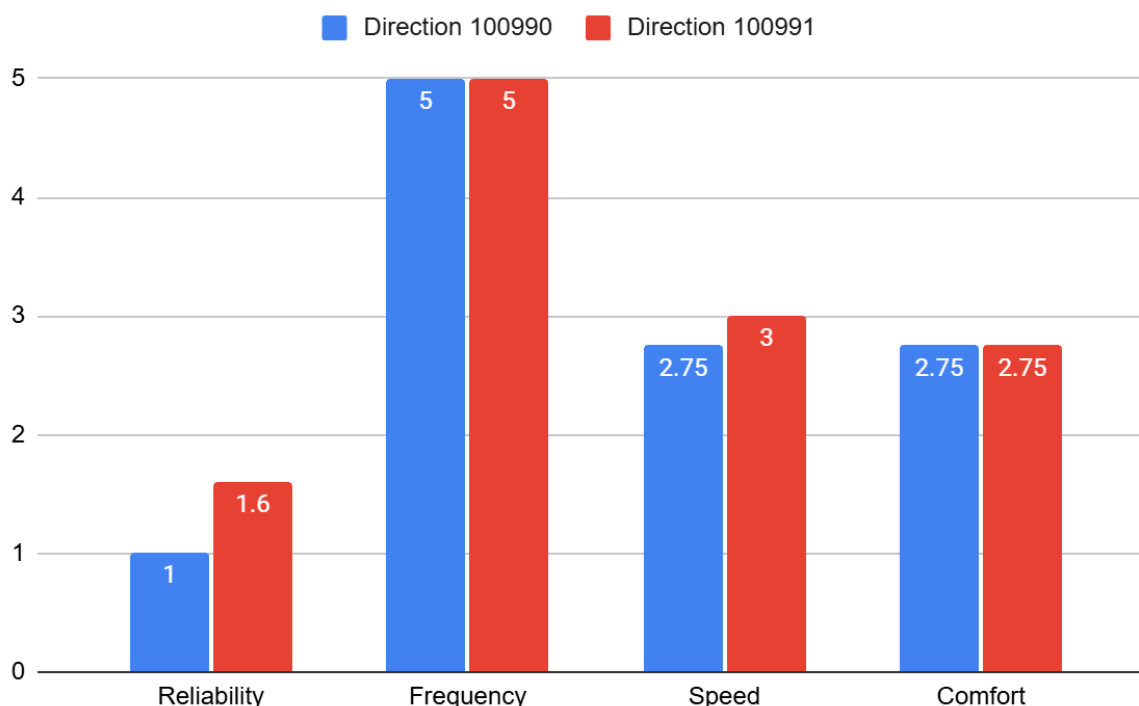
##### For Direction 100991:

$$(0.30 \times 1.60) + (0.30 \times 5.00) + (0.20 \times 3.00) + (0.20 \times 2.75) = 3.13$$

Both directions therefore receive a composite LOS grade of **C**, placing Route 99 in an overall category of acceptable but flawed service quality.

| Direction | Reliability | Frequency | Speed | Comfort | Composite Score | Final LOS |
|-----------|-------------|-----------|-------|---------|-----------------|-----------|
| 100990    | 1           | 5         | 2.75  | 2.75    | 2.9             | <b>C</b>  |
| 100991    | 1.6         | 5         | 3     | 2.75    | 3.13            | <b>C</b>  |

*Chart. LOS scores for Directions 100990 and 100991.*



#### 4.6.4 Interpretation and Limitations of the Composite Result

The shared LOS C result requires careful interpretation, because the composite score conceals strong internal differences between components. In both directions, the maximum possible frequency contribution of 1.50 provides a structural floor that keeps the overall score from falling lower, despite very weak reliability performance. At the same time, the reliability component, which scores at LOS E in both directions, acts as the principal downward force on the index. This means that the composite C should not be read as evidence of balanced performance across dimensions. Rather, it reflects the coexistence of one very strong component, frequency, with one very weak component, reliability.

The 0.23-point difference between the two directions, 3.13 for Direction 100991 and 2.90 for Direction 100990, reflects the cumulative effect of Direction 100991's slightly better reliability score and its uniformly LOS C speed performance. This difference is consistent with the broader chapter findings: Direction 100991 performs somewhat better overall in the composite sense, but still remains firmly within the same letter-grade band because both directions are constrained by severe reliability problems.

A further limitation is that the composite index does not include schedule adherence as a separate weighted component. This was a deliberate methodological choice, made to avoid double-counting with the headway reliability dimension, since both measures reflect aspects of

the same underlying service instability. However, it also means that the severe stop-level on-time performance failures identified in Section 4.2 contribute to the composite only indirectly, through their connection to headway irregularity. To that extent, the final LOS C result should be understood as a useful aggregate summary, but not as a complete substitute for the more detailed operational diagnosis developed across Sections 4.2 through 4.5.

## 5.0 Conclusions and Recommendations

### 5.1 Overview

This study evaluated the operational performance of Amman BRT Route 99 using a TCQSM-based Level of Service framework applied to one week of stop-level ITS data. Chapter 4 examined the corridor across five dimensions in sequence — schedule adherence, headway reliability and passenger waiting time, dwell time, operational efficiency and passenger comfort, and a composite LOS index. Read individually, each measure identified particular strengths and weaknesses. Read together, they describe a corridor whose problems are neither random nor evenly distributed, but follow a coherent pattern rooted in a single operational behavior and expressed differently in each direction of travel. This chapter draws those findings into an integrated diagnosis, sets out recommendations differentiated by direction and by stop type, reflects on the composite index as an evaluation tool, considers what the results imply for the wider Amman BRT network, and closes with the study's limitations and directions for future work.

### 5.2 Integrated Diagnosis: One Mismatch, Two Failure Modes

The central conclusion of this study is that the most consequential performance problems on Route 99 originate in a mismatch between how the timetable is constructed and how the route is actually operated. Buses on the corridor do not dwell to schedule: drivers stop only for active boarding and alighting and otherwise keep moving, with no holding at timing points to absorb accumulated lead. This practice, identified at the outset of Chapter 4, is not in itself a defect — it prioritizes through-trip speed and reduces in-vehicle delay — but it interacts with a static, uniformly structured schedule in two distinct ways depending on direction and time of day.

In the Outbound direction, scheduled dwell allowances exceed the dwell time actually incurred at most stops, and the surplus compounds along the route. The result is progressive early running across most of the service day, with on-time performance collapsing to LOS E and F over the majority of the corridor beyond the first few stops. In the Inbound direction, the same operating practice meets the opposite constraint during the PM peak: evening congestion pushes actual running time above the value embedded in the block, and because the schedule contains no mechanism to absorb the excess, the result is sustained and growing lateness through the second half of the route. One mechanism — a schedule that does not describe how the corridor is actually run — therefore surfaces as two different failure modes under different operating conditions.

The headway results show how this instability reaches the passenger. Route 99 is not under-supplied: mean headways were short enough to grade LOS A at every stop, in both directions, in all four periods. Yet that frequency advantage is progressively dissipated as buses move downstream. The coefficient of variation of headways rises steadily along the corridor in every period, reaching LOS F across large portions of the route and, in the worst cases, exceeding 1.0 — a condition in which variation in spacing matches or exceeds the spacing

itself. Expected waiting time degrades in step, falling to LOS D or E at the downstream end during the most affected periods. The corridor delivers frequent service but not regular service, and the practical value of its strong frequency is only partially realized past the mid-route segment.

The dwell-time analysis locates much of the source of this instability at specific stops and, importantly, distinguishes its causes. The two directions have different dwell structures. Outbound is anchored by one dominant mixed-cause bottleneck at Sport City Terminal, supported by a downstream cluster (King Abdullah Gardens through Mecca Street) where elevated dwell is only weakly related to boarding activity and therefore points to operational rather than demand-driven causes. Inbound carries a heavier and more distributed burden: two paired middle-route bottlenecks of near-equal magnitude, of which one (Stop 15) is almost entirely process-driven and the other (Stop 13) mixed-cause, together with a recurring secondary bottleneck and broader, more extreme stop-time variability — including several stops where dwell behavior is close to random. This distributed burden helps explain why Inbound exhibits more persistent reliability degradation across the corridor, while Outbound deteriorates more sharply toward its downstream end.

Finally, the efficiency and comfort results show that these reliability problems play out on a route operating close to its capacity under constrained speed. Both directions cross into LOS D loading during the PM peak — Outbound more severely, at an average cumulative load factor of 93.6% — while Inbound sustains elevated loading across most of the service day rather than only at the peak. Operating speed is consistently rather than occasionally low in both directions, reflecting the combined effect of traffic conditions and the stop-level dwell burden. The corridor therefore operates with limited recovery margin: trip times are stable, but there is little slack available to absorb disruption once delays begin to accumulate, which is precisely the condition under which dwell variability propagates most readily into headway instability.

The overall diagnosis is consequently one of an infrastructure–operation gap, as anticipated in Chapter 1. Route 99 possesses the physical and supply-side foundations of high-quality BRT — high frequency, substantial demand, stable trip durations — but its service quality is eroded at the operational layer by a schedule that does not match observed behavior, by uncontrolled vehicle spacing, and by stop-level friction that is partly demand-driven and partly procedural.

### **5.3 Recommendations**

The diagnosis points to four interventions, set out here in order of expected leverage. They are closely linked: the first corrects the schedule at its source, the next two remove the operational causes of headway irregularity, and the last addresses the capacity and comfort pressures that persist independently of reliability. It is worth noting at the outset what these recommendations do not include. The control-based strategies most prominent in the bunching literature reviewed in Chapter 2 rely on holding vehicles at timing points to restore even spacing. On a corridor that operates largely in dedicated right-of-way, deliberately delaying buses in this way would impose a direct time penalty on through passengers and would work against the purpose of the infrastructure. This study therefore treats irregular spacing not as a condition to be managed by

intervention at timing points, but as a symptom of stop-level and signal delay — and accordingly directs the recommendations at those underlying causes.

### 5.3.1 Period- and stop-specific schedule recalibration

The most consequential intervention is also among the least capital-intensive: rebuilding the timetable so that it describes how the route actually runs. The schedule adherence results are consistent with a single running-time profile applied uniformly across the service day, whose consequences then differ by direction and period — generous relative to observed conditions for most of the Outbound day, and insufficient for the Inbound PM peak. The corrective is to derive block times from the observed distributions of running time and dwell time at each stop, computed separately for each direction and each operating period and averaged across a sufficient span of days to smooth out day-to-day noise. In practical terms this means tightening Outbound block times outside the PM peak, where the current schedule embeds dwell allowance that drivers do not use, and extending Inbound PM peak block times to cover the congestion-driven running time that the current schedule fails to absorb. Because the schedule adherence analysis was conducted against a normalized baseline, these findings reflect structural slack and shortfall in the blocks themselves rather than dispatch irregularity at the terminals, so they can be acted on directly through timetable design. It is worth being precise about what the data establishes here: it shows the *consequence* of a schedule that does not vary with period-specific conditions, not the planning decision that produced it. A recalibration built from period-specific observed averages corrects that consequence regardless of how the original schedule was derived.

### 5.3.2 Off-board fare collection

Recalibrating the schedule corrects systematic early running and lateness, but it does not by itself even out spacing, because headway irregularity grows along the route through the interaction of dwell time and passenger accumulation. The first operational lever against that growth is boarding speed. At the demand-sensitive stops — the university-adjacent stops in Outbound and the strongly boarding-correlated stops in Inbound — the dwell analysis showed a clear link between boarding activity and stop time, and on-board fare payment is a direct contributor to that per-passenger boarding time. Moving fare validation off the vehicle removes the payment step from the boarding process, which reduces both the average dwell at these stops and, more importantly for reliability, its variability. Since inconsistent stop times are the input that feeds spacing irregularity downstream, cutting the variance of boarding time attacks bunching at one of its sources rather than at its symptom. A phased rollout is realistic: prioritizing the highest-volume boarding stations first — the university-adjacent stops and Sport City Terminal — would capture most of the benefit early, with extension to the remaining stops over time.

### 5.3.3 Transit signal priority

The second operational lever addresses the delay that is not related to passengers at all. Across the downstream Outbound cluster and the process-driven Inbound bottlenecks, the dwell

analysis found substantial stop time coexisting with a weak or near-zero relationship to boarding activity — delay that persists whether or not passengers are present. On Route 99, a plausible and locally recognizable component of this is signal delay at stops located at or adjacent to signalized intersections, where buses stop for the traffic phase independently of boarding demand and currently receive no priority. The consistently low operating speed across all four periods, between 21.6 and 24.6 km/h in Outbound, is consistent with the same source of delay accumulating along the corridor. Transit signal priority — extending green or shortening red for approaching buses at these intersections — would reduce this non-boarding delay directly, improve operating speed, and, by removing an erratic source of stop-to-stop delay, contribute to more even spacing. It should be noted that the ITS dwell measure captures time stopped at a station, so it evidences signal delay most clearly where a light sits at the stop itself; the case for signal priority therefore rests on the combination of the low boarding–dwell correlation, the persistently suppressed operating speed, and direct knowledge of the corridor's intersection layout, rather than on the dwell data in isolation.

### **5.3.4 Fleet standardization on high-capacity vehicles**

The final recommendation concerns capacity and comfort. Route 99 is operated with a mixed fleet: the Kent C and Kinglong vehicles carry a rated capacity of 62, while the Vectio C carries only 44, and the smaller vehicle accounts for roughly a fifth to a quarter of trips in each direction. Because load factor in this study was computed against each vehicle's own capacity, a Vectio trip carrying the same number of passengers as a larger bus registers a higher load factor purely because its denominator is smaller. Part of the PM peak crossing into LOS D, and part of the trip-to-trip variability in loading documented in Chapter 4, therefore reflects fleet heterogeneity rather than demand variation alone. Standardizing the corridor on high-capacity vehicles would accomplish two things at once: it would raise effective capacity in the periods where loading is highest — most acutely the Outbound PM peak at 93.6% — and it would remove one structural source of the crowding variability that makes the passenger experience uneven from one trip to the next. This is a more precise case than a general observation that smaller buses fill faster, and it connects directly to figures already established in the analysis.

Taken together, these four levers form a coherent program rather than a list of independent fixes. Recalibration removes the systematic schedule error; off-board fare collection and signal priority remove the two operational sources of stop-level and spacing delay; and fleet standardization protects comfort where demand is highest. Headway regularity, which appeared throughout Chapter 4 as the corridor's most damaging passenger-facing weakness, is not addressed by a separate control measure in this scheme but emerges as the expected outcome of the second and third recommendations — the result of buses that move more evenly because the delays that pull them out of spacing have been reduced.

## **5.4 The Composite Index as an Evaluation Tool**

The composite assessment placed both directions at LOS C, with scores of 2.90 Outbound and 3.13 Inbound. As a summary device, this result is communicable and allows directional



comparison, but the analysis in Chapter 4 makes clear that it should be handled with care. The composite C is produced by the coexistence of one very strong component and one very weak one: the uniform LOS A frequency contributes its maximum possible weight and establishes a structural floor under the index, while reliability — the dimension passengers experience as bunching, gapping, and unpredictable waits — sits at LOS E in both directions. A reader encountering only the composite grade would conclude that Route 99 provides acceptable mid-range service, which is true in aggregate but misleading in structure.

This experience suggests two refinements for future applications of composite scoring on the network. The first is presentational: the composite grade should always be reported alongside its weakest component, so that a strong average cannot conceal a failing dimension. The second is methodological: where a single index is required, aggregation rules that penalize internal imbalance — for example, capping the composite at one grade above the worst component, or weighting components by their distance from acceptable thresholds — would better reflect the passenger experience than a purely linear weighted mean. These refinements are offered as directions for development rather than as settled method; the weighting applied in this study was itself an applied specification, and the broader point is that any composite built on a corridor with this performance structure will be sensitive to such choices. Within this study, the composite index served its intended purpose — a summary endpoint to a detailed diagnosis — and the diagnosis, not the grade, is the operative result.

## 5.5 Implications for the Wider Amman BRT Network

Although this study examined a single corridor, its principal contribution to the wider network is methodological. Three elements of the approach transfer directly. First, the schedule baseline normalization procedure separates dispatch irregularity from in-route performance, a distinction that matters wherever published timetables and actual dispatch practice diverge — a condition unlikely to be unique to Route 99. Second, the directional, time-of-day, stop-level structure of the analysis proved essential to the diagnosis: corridor-level averages would have concealed both the directional asymmetry in schedule failure and the downstream amplification of headway irregularity, and the same masking risk applies to any corridor evaluated only in aggregate. Third, the dwell decomposition — comparing dwell with and without recorded boardings and testing the boarding–dwell relationship — provides a low-cost diagnostic for distinguishing demand-driven from process-driven stop delay using data the ITS already collects, and it directly determines which category of intervention is appropriate at each location.

The substantive findings also carry a transferable caution. Route 99 demonstrates that high service frequency, the dimension most visible in supply-side planning, can coexist with severe reliability failure. Network-level performance monitoring that tracks frequency and schedule adherence alone would rate this corridor far more favorably than its passengers experience it. Incorporating headway regularity and expected waiting time into routine monitoring — both computable from existing ITS data — would give the network a passenger-facing reliability measure that the current indicators lack.

## 5.6 Limitations and Future Work

The conclusions above should be read against the limitations set out in Chapter 3. The ITS dataset records boardings but not alightings, so the load factor results are trip-level utilization proxies rather than true onboard occupancy, and they understate peak segment loads. The seven-day observation window supports a detailed short-term diagnosis but cannot separate the structural patterns identified here from seasonal or longer-cycle variation, and the August study period coincided with the summer academic semester, meaning demand — particularly at university-adjacent stops — was an intermediate rather than maximum condition for the corridor. The on-time performance thresholds were drawn from the TCQSM 2nd Edition, and the composite weighting is an applied specification rather than a prescribed standard. None of these constraints undermines the central diagnosis, which rests on consistent and mutually reinforcing patterns across independent measures, but they bound the precision of the quantitative results and the strength of the comfort conclusions in particular.

Future work could extend the analysis in three directions. A longer observation horizon, spanning academic-term conditions, would test whether the patterns documented here persist under full demand and would allow seasonal variation to be separated from structural behavior. Alighting data, where it becomes available, would convert the load factor proxies into segment-level occupancy profiles and substantially sharpen the crowding analysis. Most directly, a controlled before-and-after evaluation of a recalibrated schedule on Route 99 would test the central claim of this study: whether correcting the schedule–operations mismatch produces the predicted improvements in schedule adherence and, through reduced irregularity at the source, in downstream headway reliability.

## 5.7 Concluding Remarks

Route 99 is not a failing service. It operates at high frequency throughout the day, carries substantial and sustained demand, and produces stable trip times in most periods. Its weaknesses — which appear at first as separate failures in schedule adherence, headway reliability, dwell behavior, and speed — are better understood as different expressions of a single mismatch between how the route is scheduled and how it is operated, presenting one face in the Outbound direction and another in the Inbound. Recognizing that asymmetry is the key to intervening effectively, because it implies that the corridor's problems are specific, locatable, and addressable rather than systemic. The most consequential improvement is also among the least costly: rebuilding the timetable around the route's observed behavior, direction by direction and period by period. Supported by off-board fare collection and signal priority at the stops where time is actually being lost, and by a uniform high-capacity fleet, these measures would let buses move more evenly through the corridor of their own accord — and a service that already delivers frequency at LOS A would finally give passengers a reason to rely on it.